

Working with nflverse Data in Python

Every NFL play since 1999 is a free download. Learn to load it, read it, filter it honestly, join the charting data to it, and rebuild this book's charts yourself.

Super Bowl LX, early in the second quarter.¹ Seattle has second-and-10 at its own 24, and Kenneth Walker III takes the handoff around the left end for 30 yards before he is pushed out of bounds at the New England 46. On the broadcast it is a highlight. By the next morning it is also a line of data: (14:04) (Shotgun) 9-K.Walker left end pushed ob at NE 46 for 30 yards (25-M.Jones) ., plus 371 other columns that record the down, the distance, the yard line, the expected points before and after, Seattle's chance of winning, who carried and who made the tackle. Every chart in this book was built from lines like that one. This chapter teaches you to load them, read them, filter them, join them to the charting data that says who was on the field, and rebuild the charts you have already seen. Along the way you'll find out why two respectable sources disagree about whether Seattle's quarterback was even in the shotgun on that play.

You'll be able to:

- Install `nflreadpy`, load a season of play-by-play in two lines, and move between Polars and pandas when you need to.
- Load schedules, rosters, participation and FTN charting data, and say what each one adds and which seasons it covers.
- Read a play-by-play row: find the situation (`down`, `ydstogo`, `yardline_100`), the result, and the model columns (`ep`, `epa`, `wp`).
- Filter to “real” plays (runs and dropbacks, meaning pass plays including sacks and scrambles, without plays wiped out by penalties, kneel-downs or garbage time) and state exactly which filter you used, and why.
- Join participation and FTN charting data to play-by-play for personnel and formation, without being fooled by the 2022-to-2023 source break.
- Reproduce a chart from an earlier chapter end to end, and score your prediction log against the new data every week.

i You'll need

- **Downs, Distance, and Field Position:** `down` and `distance`, the yard line, **expected points** (what a situation is worth on average), **EPA** (how much one play changed it) and **success rate** (the share of plays with positive EPA). This chapter shows where those numbers live in the data.
- **Personnel: Who's on the Field Before Anything Else:** **personnel groupings**, the two-digit code for backs and tight ends (**11 personnel** is one back, one tight end, three receivers). We rebuild that chapter's season-by-season chart.
- **Python:** you should be comfortable with pandas (`groupby`, `merge`, boolean filters). Nothing else is assumed.

How to read this chapter

Part 13 shows its code; the other parts fold it away. Read the cells as part of the text, and if you can, run them as you go: each cell builds on the ones above it, top to bottom, the way a notebook does.

The cells use two libraries. **nflreadpy** is nflverse's own loader, introduced below. **gridiron** is this book's small library: the diagrams, the chart style, and a thin wrapper that caches nflverse downloads. It is not on PyPI; it lives in the book's public repository, and one command installs it with everything it needs (`nflreadpy`, `Polars`, `pandas`, `matplotlib`):

```
# In a terminal:
# pip install "git+https://github.com/joshuawjulian/reading-the-game"
# Or skip gridiron entirely: every gridiron.data call below has a one-line nflreadpy
# equivalent, shown where it first appears.
```

The first cell imports what the chapter needs. Two small presentation helpers (one prints a data frame as a table, one tidies the field diagrams) are defined with the drawing code further down, folded away on the web and left out of print. The football starts in the next section.

¹Super Bowl LX, February 8, 2026, Levi's Stadium: Seattle 29, New England 13; Kenneth Walker III was the game's MVP with 27 carries for 135 yards (https://en.wikipedia.org/wiki/Super_Bowl_LX; course fact sheet). The play is nflverse game 2025_22_SEA_NE, play 1015.

```

import os
import re
import shutil

import numpy as np
import pandas as pd
import polars as pl
import matplotlib.pyplot as plt
from matplotlib.colors import LinearSegmentedColormap
from matplotlib.patches import Circle, FancyArrowPatch, FancyBboxPatch, Rectangle
from IPython.display import HTML, Markdown, display

from gridiron import Play, Field, offense, defense, style, show_animation
from gridiron.data import nfl, pbp

style.apply()

```

nflverse: free football data, and why it exists

nflverse is a volunteer, open-source project that publishes free, cleaned NFL data and the packages that load it. Its centrepiece is **play-by-play** data, usually shortened to **pbp**: one row for every play of every game since the 1999 season, about 48,000 rows a season, each with a few hundred columns.

Where does it come from? The raw material is the league’s own play-by-play record of each game, which nflfastR reads from the NFL’s game-data feeds: the play descriptions entered by each game’s official statistics crew, the same text printed in the gamebook a reporter picks up after the game.² That record says what happened in words and a handful of fields. nflverse turns it into a table, fixes its inconsistencies (team abbreviations that changed when franchises moved, plays logged out of order, penalties enforced from odd spots) and then adds the columns that make it useful for analysis: who was the passer and the receiver, expected points and EPA, win probability, completion probability and more. Those model columns are the reason the data matters. Anyone can count yards; EPA on every play since 1999, free, is what let public analysts argue with professional ones.

The project has a short, busy history. It began at Carnegie Mellon, where the statisticians Maksim Horowitz, Ronald Yurko and Samuel Ventura built **nflscrapR**, an R package that scraped the league’s game data and fitted the first widely used open-source expected-points and win-probability models.³ In April 2020 Ben Baldwin and Sebastian Carl released **nflfastR**, a much faster rebuild with new models that now reaches back to 1999, and the family of packages that grew up around it took the name nflverse.⁴ None of these people built it for a team. The data exists because a handful of people with statistics training, who also happened to love football, decided it should be free for anyone.

For Python users, the loader is **nflreadpy**: one function per table, each returning a Polars data frame. It replaced **nfl_data_py**, the older Python package, which is now archived; its own README says it is deprecated in favour of nflreadpy and will get no further maintenance.⁵ In October 2026 the current release of nflreadpy is 0.1.5 (November 2025), and the repository is still active.⁶

²nflfastR’s CRAN description (“functions to efficiently access NFL play-by-play data from nfl.com”): <https://cran.r-project.org/package=nflfastR>; its README credits the JSON-formatted NFL play-by-play feeds back to 1999 and notes that the hosted data “are updated nightly during the season”: <https://github.com/nflverse/nflfastR>.

³Maksim Horowitz, Ronald Yurko and Samuel Ventura, nflscrapR (R package): <https://github.com/maksimhorowitz/nflscrapR>; the expected-points and win-probability models are described in Ronald Yurko, Samuel Ventura and Maksim Horowitz, “nflWAR: A reproducible method for offensive player evaluation in football,” *Journal of Quantitative Analysis in Sports* 15, no. 3 (2019): 163-183, <https://doi.org/10.1515/jqas-2018-0010>.

⁴nflfastR documentation (authors Sebastian Carl and Ben Baldwin): <https://www.nflfastR.com/>; nflfastR was announced by Ben Baldwin on April 27, 2020 (tweet embedded in Thomas Mock, “nflfastR + dbplyr + RSQLite,” The Mockup blog: <https://themockup.blog/posts/2019-04-28-nflfastR-dplyr-rsqlite/>), and first published on CRAN on September 1, 2020 (<https://cran.r-project.org/package=nflfastR>); Ben Baldwin, “nflfastR EP, WP, and CP models,” Open Source Football, September 28, 2020: <https://www.opensourcefootball.com/posts/2020-09-28-nflfastR-ep-wp-and-cp-models/>; the nflverse organisation: <https://github.com/nflverse>.

⁵nflverse, nfl_data_py (archived repository; README: deprecated in favour of nflreadpy, no further maintenance): https://github.com/nflverse/nfl_data_py.

⁶nflverse, nflreadpy: <https://github.com/nflverse/nflreadpy>; version 0.1.5 released on PyPI on November 19, 2025 (<https://pypi.org/project/nflreadpy/>), per the course fact sheet, checked October 2026.

Drawing code (boxes, arrows, label positions, font sizes) is about matplotlib, not football or data, so this chapter keeps it in helper functions that are folded away in the web version (“Show the plotting code”) and left out of the printed one. Every line that loads, filters, joins or counts data is shown; each chart cell ends by handing its numbers to a `plot_...` function.

Here is how the pieces fit together, from the stadium to the pictures in this book:

```
draw_flow()
plt.show()
```

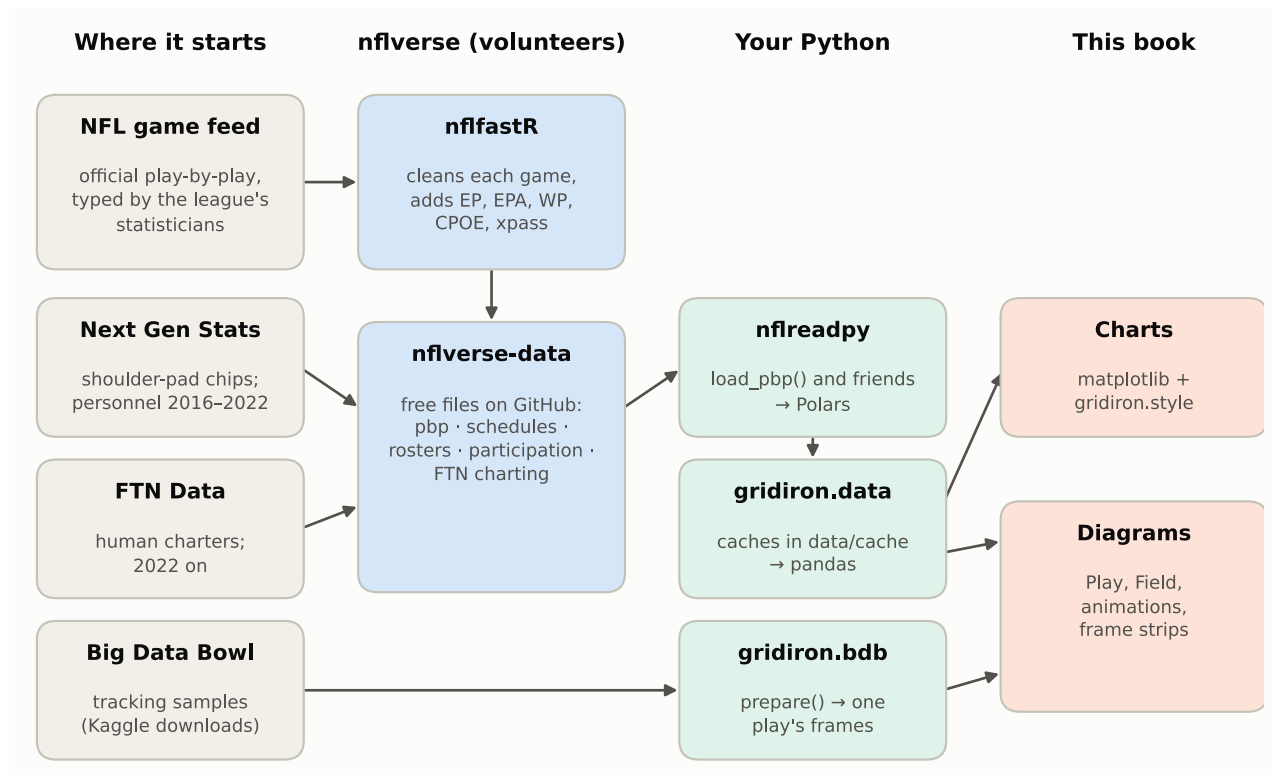


Figure 1: Where this book’s numbers and pictures come from. What to notice: nothing in the nflverse column records football itself; nflverse republishes three suppliers’ records and adds its own models. Left: the three suppliers (the league’s official play-by-play, Next Gen Stats tracking from chips in the players’ shoulder pads, and FTN’s human charters), plus the Big Data Bowl’s tracking samples, which you download from Kaggle yourself. nflfastR turns the league’s record into a clean table and adds the model columns; nflverse publishes everything as free files. In Python, nflreadpy downloads them as Polars data frames, and the course’s gridiron.data wrapper caches them and hands over pandas. Charts use gridiron.style; diagrams and animations use gridiron’s Play and Field, fed either by a hand-drawn play or by real tracking frames through gridiron.bdb.

Hold on to the shape of that picture. nflverse is a **republisher**, not a source: it does not chart games itself. The play-by-play reflects what the league’s statisticians wrote. The personnel and formation data reflect what Next Gen Stats (the league’s tracking system, which follows radio chips in every player’s shoulder pads; see Tracking Data and the Big Data Bowl) or FTN’s charters recorded. When those suppliers make a mistake, or change how they record something, nflverse inherits it. The **models**, on the other hand, are nflverse’s own. When this book quotes EPA or win probability, it is quoting nflfastR’s models, which are good, public and documented, but are one model among several. ESPN’s win probability on a broadcast is a different model and will not match to the decimal.

! Why it exists

Before free play-by-play, serious football analysis lived inside teams and a few paid services. You could read that “teams pass too much on first down” but you could not check it. A public, reproducible table changes the argument: anyone can run the numbers, find the flaw in someone else’s, and publish a better version. The cost is that you inherit the data’s quirks, and this chapter is mostly about those quirks. The trade is worth it.

Installing nflreadpy and loading your first season

On your own machine, the whole setup is one install and two lines of Python:

```
# In a terminal:  pip install nflreadpy pandas pyarrow      (or: uv add nflreadpy pandas pyarrow)
import nflreadpy as nfl

plays = nfl.load_pbp([2025])      # a Polars DataFrame, one row per play
plays_pd = plays.to_pandas()      # the same data as pandas (needs pyarrow)
```

The first call downloads the season from nflverse’s GitHub releases, a Parquet file of about 20 megabytes, so it takes a few seconds. Every loader takes a list of seasons, a single season, or `True` for everything available.

This book wraps nflreadpy in a tiny module, `gridiron.data`, so that every chapter shares one cache on disk. `nfl()` returns the nflreadpy module, already pointed at that cache, and `pbp(seasons, columns=[...])` loads seasons, keeps the columns you name and hands back pandas. Without gridiron, the plain nflreadpy versions are:

```
import nflreadpy as nfl          # instead of  nflr = nfl()
hist = nfl.load_pbp(seasons).select(cols).to_pandas()  # instead of  pbp(seasons, columns=cols)
```

```
nflr = nfl()                     # the nflreadpy module, pointed at data/cache
raw25 = nflr.load_pbp([2025])    # Polars
print(type(raw25).__name__, f"{raw25.height:,} rows x {raw25.width} columns")
```

```
DataFrame 48,771 rows x 372 columns
```

Polars or pandas?

nflreadpy returns **Polars** data frames. Polars is a newer data-frame library, written in Rust, that stores each column as one block of memory and runs operations in parallel; on a ten-season load it is noticeably faster than pandas and uses less memory. Most Python tutorials, and most of this book, use pandas, because it is what most readers know. You can work in either. Here is the same question, “how often did offenses pass on each down in the 2025 regular season?”, asked both ways:

```
# Polars: chain expressions on the frame nflreadpy returned
by_down_pl = (
    raw25.filter((pl.col("season_type") == "REG")
                 & pl.col("play_type").isin(["pass", "run"])
                 & pl.col("down").is_not_null())
    .group_by("down")
    .agg(pl.col("pass").mean().alias("pass_rate"), pl.len().alias("plays"))
    .sort("down")
)
print(by_down_pl)

# pandas: convert once, then the familiar idioms
df25 = raw25.to_pandas()
reg25 = df25[(df25.season_type == "REG") & df25.play_type.isin(["pass", "run"])
             & df25.down.notna()]
print(reg25.groupby("down")["pass"].agg(pass_rate="mean", plays="size").round(3))
```

```
shape: (4, 3)
```

down	pass_rate	plays
---	---	---

f64	f64	u32
1.0	0.497722	14267
2.0	0.617417	10863
3.0	0.785032	6801
4.0	0.6678	882

```

pass_rate plays
down
1.0      0.498 14267
2.0      0.617 10863
3.0      0.785  6801
4.0      0.668   882

```

The answers are identical, as they should be. To make the conversion painless, convert **after** you have selected the columns you need (`raw25.select(...).to_pandas()`), because a 372-column frame is mostly columns you will never touch; `gridiron.data.pbp(seasons, columns=...)` does exactly that. And remember that Polars has no index: if pandas code does `.loc[...]` on the result of a groupby, it is relying on the index pandas created, which Polars never had.

The cache, and the current season

nflreadpy keeps what it downloads. By default the cache lives **in memory** for 24 hours, so a second load in the same session is instant and a new session downloads again. The course instead caches **on disk** for a year, so the book can be rebuilt without fetching ten seasons every time:

```

from nflreadpy.config import update_config

update_config(cache_mode="filesystem", cache_dir="data/cache",
              cache_duration=60 * 60 * 24 * 365)  # seconds; the course keeps files a year

```

A year is right for completed seasons, which change rarely. It is wrong for the season in progress. nflverse refreshes the current season's play-by-play as games are played, so during the season you want a short cache (say `cache_duration=6 * 60 * 60`, six hours) or a fresh download (`nflreadpy.clear_cache()`). The Sunday notebook at the end of the chapter does exactly that.

Common misconception

“The Python package for NFL data is `nfl_data_py`.” It was, and thousands of tutorials and notebooks still use it (`import nfl_data_py as nfl; nfl.import_pbp_data([2023])`). It is now archived and unmaintained, so it will drift away from the data as nflverse changes its files. Use nflreadpy: the function names start with `load_` instead of `import_`, it returns Polars instead of pandas, and otherwise the data is the same nflverse data.

Anatomy of a play-by-play row

Here is Walker's run as nflverse records it, a handful of its 372 columns:

```

walker = df25[(df25.game_id == "2025_22_SEA_NE") & (df25.play_id == 1015)].iloc[0]
MEANING = {"game_id": "season _ week _ away team _ home team", "play_id": "the play's number in the game",
           "qtr": "quarter", "time": "game clock at the snap", "posteam": "team with the ball",
           "down": "down", "ydstogo": "yards to the line to gain",
           "yardline_100": "yards to the opponent's goal line", "play_type": "kind of play",
           "yards_gained": "yards gained on the play", "ep": "expected points before the snap",
           "epa": "expected points added by the play",
           "wp": "offense's win probability before the snap"}
FMT = {"play_id": "{:.0f}", "qtr": "{:.0f}", "down": "{:.0f}", "ydstogo": "{:.0f}",
       "yardline_100": "{:.0f}", "yards_gained": "{:.0f}", "ep": "{:.2f}", "epa": "{:.2f}",

```

```

"wp": "{:.2f}"
md_table(pd.DataFrame({"column": list(MEANING),
                        "value": [FMT.get(c, "{}").format(walker[c]) for c in MEANING],
                        "meaning": list(MEANING.values())}))

```

column	value	meaning
game_id	2025_22_SEA_NE	season _ week _ away team _ home team
play_id	1015	the play's number in the game
qtr	2	quarter
time	14:04	game clock at the snap
posteam	SEA	team with the ball
down	2	down
ydstogo	10	yards to the line to gain
yardline_100	76	yards to the opponent's goal line
play_type	run	kind of play
yards_gained	30	yards gained on the play
ep	0.12	expected points before the snap
epa	+2.62	expected points added by the play
wp	0.53	offense's win probability before the snap

Read it left to right and it is a scorebug plus a result. `game_id` names the game (season, week, away team, home team: Seattle was the designated away team), and `play_id` numbers the play within it. Seattle (`posteam`, the team with the ball) has second down (`down`) and 10 to go (`ydstogo`) at its own 24, which `nflverse` writes as `yardline_100 = 76`; the next section explains why. The play was a run that gained 30 yards. Before the snap the situation was worth 0.12 expected points (`ep`); the run added 2.62 (`epa`), because it turned second-and-10 deep in Seattle's half into first-and-10 in New England's. Before the snap, `nflfastR`'s **win probability** model gave Seattle a 53% chance of winning (`wp`), its estimate from the score, the clock, the down, distance and field position, and the timeouts left. It deliberately ignores the betting line; a sister column, `vegas_wp`, folds in the pre-game point spread and, because Seattle was favoured, gives it 72%.

The other 359 columns fall into a few families. You will use perhaps forty of them regularly, so it pays to know which family to look in. The book sorts them with a few name patterns (folded below; the sorting is the book's, not `nflverse`'s):

```

fam = pd.Series({c: family(c) for c in raw25.columns})
counts = fam.value_counts()
print(counts.to_string())
draw_schema(schema_examples(walker), counts, walker.desc)
plt.show()

```

What happened	125
Players	102
Model estimates	64
Drive and game	46
Situation before the snap	24
Keys	11



Figure 2: The 372 columns of a 2025 play-by-play row, sorted into six families (counts from the code; the sorting is the book's, not nflverse's), with Walker's 30-yard run from Super Bowl LX as the example row. Keys identify the play; the situation is what the scorebug showed before the snap; what happened is the result; the model columns are nflfastR's estimates; the player columns name and identify everyone involved; the drive and game columns describe the possession and the game around it. The play description at the bottom is the original text the league's statisticians wrote.

The **situation** columns are the scorebug, so they are what you filter on. The **model** columns are where the analysis is, and each one has a plain meaning:

- **ep** and **epa**: expected points before the snap, and how much the play changed them. **success** is 1 when **epa** is positive, so its mean is the success rate.
- **wp** and **wpa**: win probability before the snap, and *win probability added*, the change the play caused. Walker's run moved Seattle's chance of winning by +7.7 points (**wpa** = +0.077).
- **cp** and **cpoe**, on passes only: completion probability, the model's chance that a throw of that depth and direction in that situation is caught, and completion percentage over expected, caught (1 or 0) minus **cp**.
- **xpass** and **pass_oe**: the model's chance that an average offense drops back in this situation, and pass over expected, the play's **pass** flag minus **xpass**, in percentage points. Second-and-10 at your own 24 has **xpass** = 0.61: a 61% chance of a dropback. Seattle ran, so **pass_oe** = -61. Averaged over a team's snaps, that is PROE.

They are estimates, not facts. Expected Points, EPA, and Success Rate builds the first of those models from scratch, Win Probability and Decision Analysis the second, and Tendencies: PROE, Personnel, Motion, and Coverage the last.

Here are the columns you will reach for most, by job. Keep this table open while you work:

Job	Columns
Identify the play	game_id, play_id, season, week, season_type, posteam, defteam
The situation	down, ydstogo, yardline_100, qtr, half_seconds_remaining, score_differential, goal_to_go

Job	Columns
What kind of play	play_type, pass, rush, qb_dropback, qb_scramble, sack, shotgun, no_huddle, qb_kneel, qb_spike, two_point_attempt
The result	yards_gained, first_down, touchdown, interception, fumble_lost, complete_pass, air_yards, yards_after_catch, pass_location, pass_length, run_location, run_gap, penalty, penalty_type
The value	ep, epa, success, wp, wpa, cpoe, xpass, pass_oe
Who	passer_player_name, rusher_player_name, receiver_player_name, passer_id, rusher_id, receiver_id
In words	desc

i Madden vs. real life

In Madden every play has a name: you pick “Inside Zone” or “PA Boot Over” from a menu, and the game knows what you called. Real play-by-play never records the call. It records what happened (“left end for 30 yards”), in the official statisticians’ words, and nothing about whether it was outside zone, a toss or a run-pass option. Even the flags that sound like scheme (`shotgun`, `no_huddle`, `run_location`) are describing the result, not reading the playbook. The scheme labels you’ll meet later in this chapter come from human charters watching film, and they are judgement calls.

A row is not always a play

The table has one row per *event* in the league’s log, and not every event is a run or a pass. Here is the 2025 file by `play_type`:

```
pt = (df25.play_type.fillna("(none)").value_counts().rename_axis("play_type")
      .reset_index(name="rows"))
pt["share"] = pt.rows / pt.rows.sum()
md_table(pt, digits=3)
```

play_type	rows	share
pass	19,737	0.405
run	14,895	0.305
no_play	4,723	0.097
kickoff	2,927	0.060
punt	2,042	0.042
(none)	1,445	0.030
extra_point	1,330	0.027
field_goal	1,140	0.023
qb_kneel	452	0.009
qb_spike	80	0.002

About 70% of the rows are runs and passes. The rest are kickoffs, punts, extra points and field goals; kneel-downs and spikes; `no_play`; and rows with no `play_type` at all, which are bookkeeping: the start of the game, the end of each quarter and of the game, and the odd weather delay. `no_play` covers three different things: a play that was run and then wiped out by an accepted penalty (a holding call, say), a pre-snap foul after which no play was run at all (a false start, the most common penalty on these rows), and a timeout. Every analysis starts by deciding which of these to keep.

yardline_100: one number for field position

yardline_100 is the number of yards from the ball to the goal line the offense is attacking. Your own 25 is 75; midfield is 50; the opponent's 10 is 10. The broadcast's way of saying it, "SEA 24," needs you to know which team has the ball and which way it is going; `yardln`, the column that stores the broadcast form, has the same problem. **yardline_100** doesn't. The figure also shows the third scale you'll meet, the Big Data Bowl's `x`, which measures the whole field from 0 to 120 including both 10-yard end zones, so the goal line the offense attacks sits at $x = 110 - \text{yardline_100}$:

```
start, gain = float(walker.yardline_100), float(walker.yards_gained)
print(f"x at the snap = 110 - {start:.0f} = {110 - start:.0f}; "
      f"x at the end = 110 - {start - gain:.0f} = {110 - (start - gain):.0f}")
plot_yardline(start, gain, float(walker.ydstogo))
plt.show()
```

x at the snap = $110 - 76 = 34$; x at the end = $110 - 46 = 64$

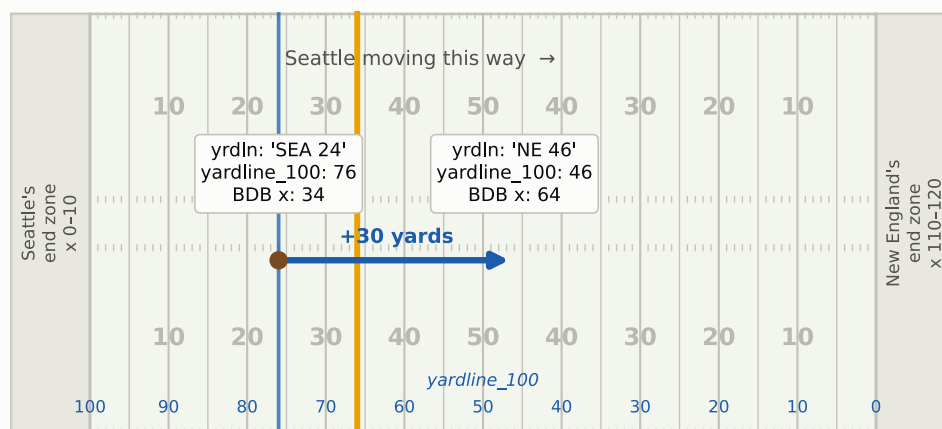


Figure 3: Walker's 30-yard run on the field, with Seattle moving left to right. The broadcast and the `yardln` column call the starting spot 'SEA 24'; `yardline_100` calls it 76, the yards still to go for a touchdown; the Big Data Bowl tracking frame (offense normalized to move toward higher `x`) puts it at $x = 34$: BDB `x` runs from 0 to 120 including both 10-yard end zones, so New England's goal line is at $x = 110$ and $x = 110 - \text{yardline_100}$. The run ends at 'NE 46': `yardline_100` = 46, $x = 64$. The yellow line is the line to gain. The bottom ruler shows `yardline_100` from 100 at Seattle's goal line to 0 at New England's.

! Why it exists

Football's spoken language for field position is relative ("own 24," "plus-46," "the 30 going in"), and the same words mean different spots for the two teams. Every computation that touches field position (expected points, red-zone filters, "how far from the end zone did this drive start?") needs a single scale that runs in the offense's direction. **yardline_100** is that scale. `yardline_100 <= 20` is the red zone for whichever team has the ball, and $110 - \text{yardline_100}$ puts the ball on the Big Data Bowl's `x` axis for an offense moving toward higher `x`, which is how this book's diagrams are drawn. The trade-off is that it is useless for "which end of the stadium?" questions (wind, sun, the home crowd), for which you need `yardln` and `posteam` together.

Flags, not categories: `pass`, `rush` and `play_type`

`play_type` sounds like the answer to "was it a run or a pass?" It isn't quite. A **scramble** is a quarterback who drops back to throw, finds no one open and runs; `nflverse` files it as `play_type == "run"`, because the ball was carried, not thrown. A **sack** is `play_type == "pass"`. The 0/1 flag columns describe the play the offense set out to run:

```
flagged = df25[df25.season_type == "REG"].copy()
flagged["kind"] = np.select( # first match wins, so no-plays are checked first
```

```

[flagged.play_type == "no_play", flagged.two_point_attempt == 1, flagged.qb_scramble == 1,
 flagged.sack == 1, flagged.qb_kneel == 1, flagged.qb_spike == 1,
 flagged.play_type == "pass", flagged.play_type == "run"],
["no-play", "two-point try", "scramble", "sack", "kneel-down", "spike",
 "pass", "designed run"], "other")
keep = ["designed run", "pass", "sack", "scramble", "two-point try", "kneel-down", "spike",
 "no-play"]
flag_tab = (flagged[flagged.kind.isin(keep)]
            .groupby("kind")
            .agg(plays=("play_id", "size"), play_type=("play_type", lambda s: s.mode()[0]),
                 has_down=("down", lambda s: s.notna().mean()), pass_flag=("pass", "mean"),
                 rush_flag=("rush", "mean"), qb_dropback=("qb_dropback", "mean"))
            .reindex(keep).reset_index())
md_table(flag_tab)

```

kind	plays	play_type	has_down	pass_flag	rush_flag	qb_dropback
designed run	13,079	run	1.00	0.00	1.00	0.00
pass	17,358	pass	1.00	1.00	0.00	1.00
sack	1,287	pass	1.00	1.00	0.00	1.00
scramble	1,089	run	1.00	1.00	0.00	1.00
two-point try	128	pass	0.00	0.76	0.24	0.75
kneel-down	433	qb_kneel	1.00	0.00	0.00	0.00
spike	79	qb_spike	1.00	0.00	0.00	0.00
no-play	4,538	no_play	0.55	0.25	0.07	0.00

Read the table row by row and the logic is clear. `pass == 1` marks every **dropback**, a play on which the quarterback set up to pass: real passes, sacks and scrambles. `rush == 1` marks only **designed runs**, including quarterback runs that were called as runs. Two-point tries have **no down**, so a `down.notna()` filter quietly drops them, which is usually what you want. Kneel-downs and spikes get their own `play_type` and neither flag. And `no_play` rows sometimes carry a flag (a run called back for holding is still marked as a run; so are the 71 scrambles wiped out by penalties, which is why they sit in the no-play row, not the scramble row) and sometimes don't (a timeout, a false start).

This is why this book counts a “pass” as `pass == 1`, and says so in its captions. It is the closest the data comes to what the offense called. Count passes with `play_type == "pass"` and you lose the scrambles, about 1,089 plays in 2025, and every pass rate comes out about three points low.

Two cautions keep the flag honest. Screens and play-action passes are passes under it, which is right for play-calling but worth remembering when a “pass rate” includes a lot of throws behind the line. And on a run-pass option (RPO) the call *was* both: the quarterback decides after the snap, and `pass` and `rush` record his decision, not the call. The only place the RPO call itself shows up is FTN's `is_rpo` flag, which you'll meet below.

Filtering to “real” plays

Almost every question in this book starts from the same funnel: the regular season, runs and dropbacks only, with a down, and (for most questions) without garbage time. Here is what each step removes from the 2025 file:

```

s0 = df25
s1 = s0[s0.season_type == "REG"]
s2 = s1[s1.play_type.isin(["pass", "run"])]
s3 = s2[s2.down.notna()]
s4 = s3[s3.wp.between(0.10, 0.90)]
s5 = s4[s4.half_seconds_remaining > 120]

```

```
plot_funnel(["Every row in the 2025 file", s0, ""),
            ("Regular season only", s1, "playoff games"),
            ("Runs and passes (play_type)", s2, "kicks, no-plays, kneels, spikes, timeouts"),
            ("...that have a down", s3, "two-point tries"),
            ("Win probability 10-90%", s4, "lopsided games"),
            ("Not in the last 2 min of a half", s5, "two-minute drills")])
plt.show()
```

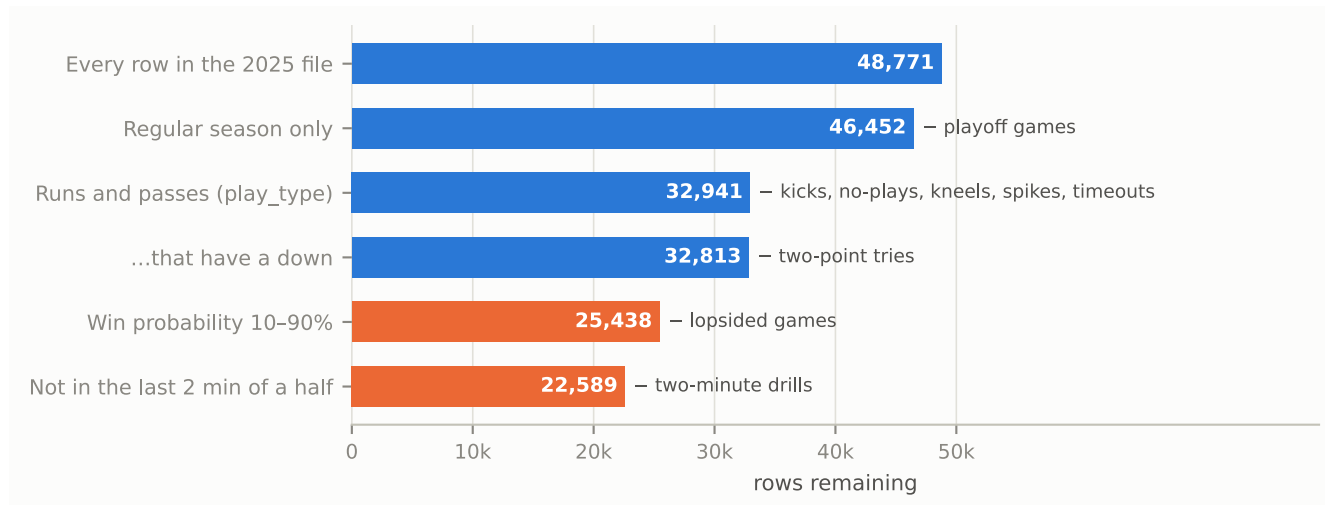


Figure 4: From every row of the 2025 file to the plays an analysis uses (nflverse play-by-play). Each bar is what survives the filter on its left; the text on the right says what that step removed. The last two steps (orange) are the garbage-time filter described below. Your own funnel may differ; the point is to know, and to say, which steps you took.

The first four steps are housekeeping, but each has a reason. **Regular season only**, because the playoffs are a different sample: 12 or 14 teams instead of 32, at most four games each, against better opponents and with elimination on the line, so mixing them in tilts any team's numbers toward how far it went. (Super Bowl LX is this chapter's example play; it is the exception, not the data.) **play_type in pass or run** may look odd after the last section warned you off `play_type`, but it is the right tool for *choosing rows*: scrambles live inside "run" and sacks inside "pass", so every dropback survives, and you then *classify* the rows with the `pass` flag. The step does drop no-play rows, including runs and passes wiped out by penalties. nflfastR's beginner's guide filters rows with `pass == 1 | rush == 1` instead, which keeps those called-back plays; Drill 3 below asks when that matters. **A down** removes two-point tries.

The last two steps are a choice, and they need explaining.

Why garbage time gets thrown out

Think about what a team does when it trails by 20 with eight minutes left. It throws on nearly every down, because runs burn clock it doesn't have, and the defense, which knows it, drops eight players into coverage and gives up underneath completions all day. Meanwhile the team that leads by 20 runs into a stacked box to keep the clock moving. Neither side is running its game plan. If you leave those plays in, a team that spent the second half of a blowout throwing will look pass-happy, and a team that spent it running out the clock will look like a running team. The score made those calls, not the coaches' preferences. It bends **efficiency** too: a trailing offense collecting soft underneath yards against a deep shell looks better than it is, and the defense that is protecting a big lead is conceding yards on purpose. Here is how strongly the score pulls on the most ordinary situation in football, first-and-10:

```
HCOLS = ["season", "season_type", "week", "game_id", "play_id", "qtr", "time", "posteam",
         "down", "ydstogo", "yardline_100", "play_type", "pass", "rush", "shotgun", "wp",
         "half_seconds_remaining", "epa", "first_down", "yards_gained", "penalty_type"]
hist = pbp(range(1999, 2026), columns=HCOLS) # 27 seasons, about 1.3 million rows
hist = hist[hist.season_type == "REG"]
snaps = hist[hist.play_type.isin(["pass", "run"]) & hist.down.notna()].copy()
```

```
f10 = snaps[snaps.season.between(2016, 2025) & (snaps.down == 1) & (snaps.ydstogo == 10)
          & (snaps.half_seconds_remaining > 120) & snaps.wp.notna()].copy()
f10["bin"] = pd.cut(f10.wp, np.linspace(0, 1, 11))          # ten win-probability bins
wp_tab = f10.groupby("bin", observed=True)["pass"].agg(["mean", "size"])
plot_wp_pass(wp_tab)
plt.show()
```

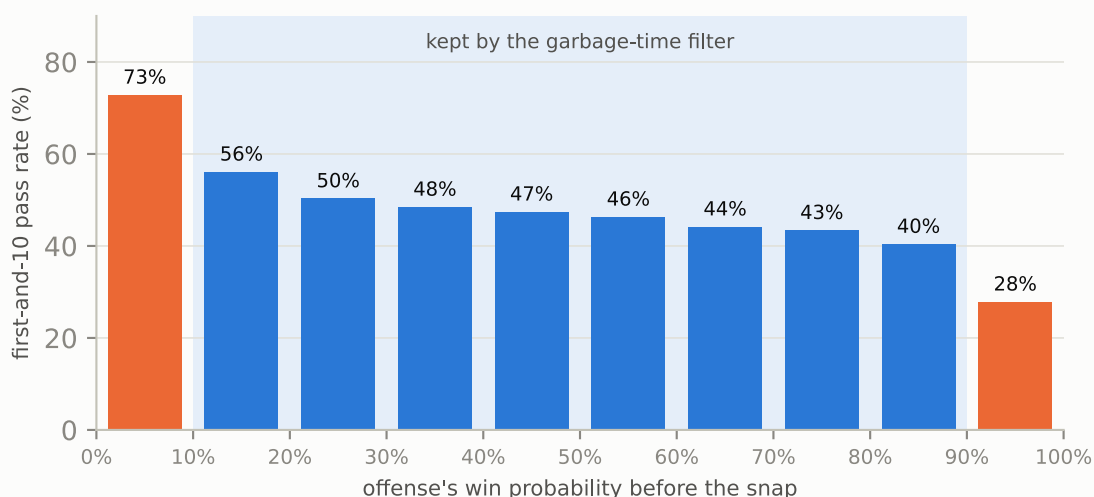


Figure 5: How often offenses passed on first-and-10, by the offense’s win probability before the snap (regular seasons 2016–2025, outside the last two minutes of each half; a pass is any dropback, including sacks and scrambles). Between about 20% and 80% the pass rate drifts only gently, from about half to a little over 40%, as teams that are ahead lean on the run. Outside that band the scoreboard takes over: teams that are almost beaten throw on about 70% of first downs, and teams that have the game won throw on fewer than three in ten. The shaded band is what this course’s garbage-time filter keeps.

The middle of the chart slopes gently, and the ends fall off a cliff. That middle is what the **garbage-time filter** tries to keep: plays where the game is still in doubt, so the call reflects the plan. This book’s version keeps plays where the offense’s win probability is **between 10% and 90%** and that are **not in the last two minutes of either half**. The win-probability part removes blowouts; the clock part removes the two-minute drill, where even a close game is played by the clock, with hurry-up passing and timeouts.

There is no single right cut-off, only a trade between keeping the plays that tell you about the game plan and keeping enough plays to measure anything. This book uses two, and here is the rule for choosing:

- **10–90% (the garbage-time filter) is the default**, and the filter for **efficiency**: EPA, success rate, yards per play, and league-wide averages over many seasons. Expected Points, EPA, and Success Rate uses it for every team number.
- **20–80% is for comparing play-callers’ choices**, where the score’s pull starts earlier and small differences between teams are the point. Tendencies: PROE, Personnel, Motion, and Coverage defines it as the **neutral situation**, usually on first and second down only. Earlier chapters used the same idea before it had a formal name: the pass-rate grid in Downs, Distance, and Field Position (20–80%, every down) and the play-calling charts in Personnel (20–80%, first and second down).

Public analysts use cut-offs in the same neighbourhood; nflfastR’s own beginner’s guide, for example, measures early-down pass rates at 20% to 80%, on first and second down in the first half, outside its last two minutes.⁷ Whatever you choose, the choice changes the numbers, so you state it. The table below compares no filter with the two win-probability ranges, each combined with the same clock rule:

```
s25 = snaps[snaps.season == 2025]
late = s25.half_seconds_remaining <= 120
CHOICES = {"none": s25,                                     # every run and dropback
           "10–90% (default)": s25[s25.wp.between(0.10, 0.90) & ~late], # garbage-time filter
```

⁷nflfastR, “A beginner’s guide to nflfastR” (its early-down pass-rate example filters on `wp > .20 & wp < .80 & down <= 2 & qtr <= 2 & half_seconds_remaining > 120`, “excluding the final 2 minutes of the half when everyone is pass-happy”; checked October 2026): https://nflfastR.com/articles/beginners_guide.html.

```

"20–80% (neutral)": s25[s25.wp.between(0.20, 0.80) & ~late]} # play-calling
choice_tab = pd.DataFrame({name: d.groupby("down")["pass"].mean() for name, d in CHOICES.items()}).T
choice_tab.columns = [f"down {int(c)}" for c in choice_tab.columns]
shown = choice_tab.map(lambda v: f"{v:.1%}")
shown.insert(0, "plays", [f"{len(d):,}" for d in CHOICES.values()])
md_table(shown.reset_index(names="WP filter"))

```

WP filter	plays	down 1	down 2	down 3	down 4
none	32,813	49.8%	61.7%	78.5%	66.8%
10–90% (default)	22,589	46.4%	59.6%	79.5%	56.8%
20–80% (neutral)	18,426	46.0%	59.4%	79.7%	54.8%

Filtering garbage time removes about three plays in ten and moves first-down pass rate by about three points, which is the size of the differences many arguments are about. Fourth down moves most, from 67% passes to 57%, because so many fourth-down plays are desperate throws by teams that are losing. The two filters agree within two points on every down; the unfiltered numbers are the odd ones out. That is why a league-wide history can use the default, while a comparison of two play-callers, where two points is the difference, should use the stricter one.

Common misconception

“Filtering is cheating: you’re throwing away data you don’t like.” Filtering is choosing the question. “How often does this team pass when the game is in doubt?” and “how often does this team pass?” are different questions, and both are legitimate. The cheating version is choosing the filter *after* seeing which one gives the answer you wanted, or not saying which you used. A filter stated in a caption is a method; one hidden in the code is a thumb on the scale.

Beyond play-by-play: schedules, rosters, participation and FTN

Play-by-play tells you what happened. It doesn’t tell you who was on the field, how they lined up, or whether the quarterback faked a handoff first. For that, nflverse publishes other tables, each with its own coverage in time. Knowing which seasons each one covers is half the battle:

```

draw_timeline()
plt.show()

```

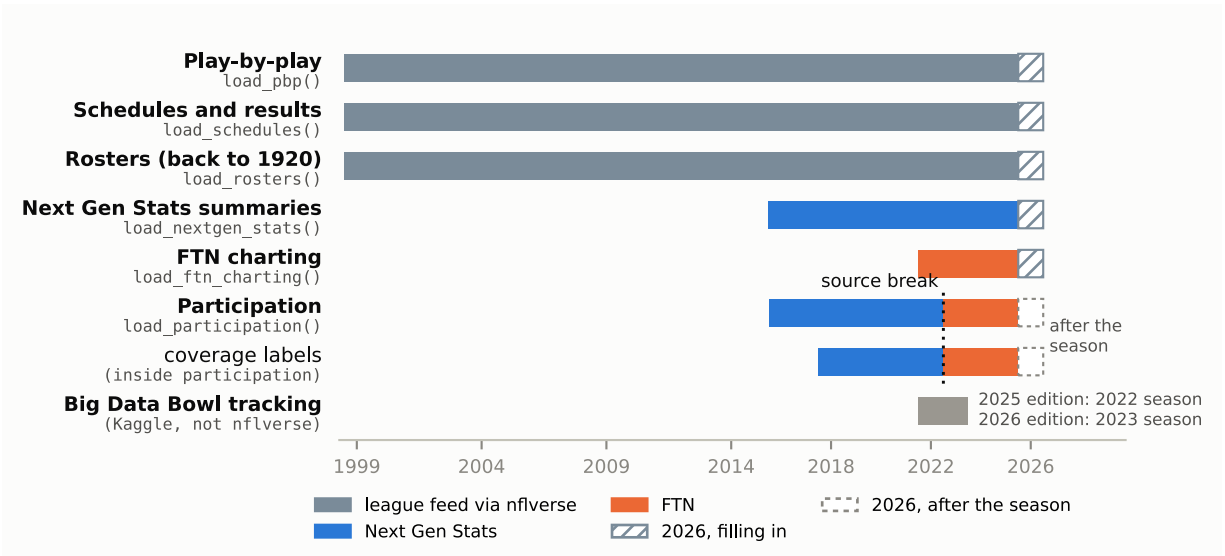


Figure 6: Which seasons each nflverse table covers, as of October 2026. Play-by-play and schedules start in 1999 (rosters go back to 1920); Next Gen Stats player summaries and participation start in 2016; FTN charting starts in 2022. Participation is supplied by Next Gen Stats (blue) through 2022 and by FTN (orange) from 2023: the dotted source break runs only through those rows, because the other tables kept one supplier. Coverage labels sit inside participation, from 2018 on. Hatched 2026 cells are being filled as the season is played; the dashed participation cells for 2026 arrive only after the season, around February 2027. Big Data Bowl tracking is not part of nflverse, and its editions are named for the year of the contest, not the season: the 2025 edition released 2022 tracking, the 2026 edition 2023 tracking.

Each table joins to play-by-play on a key, and the key is the first thing to learn about it.

One key needs a word first, because three game IDs float around. `game_id` is nflverse's readable key (2025_22_SEA_NE: season, week, away team, home team). Participation and FTN call the same key `nflverse_game_id`. `old_game_id` is the league's own ten-digit ID (2026020800, the date plus a game number), which is the one the Big Data Bowl files use. Team abbreviations are the franchise's *current* ones in every season, so `posteam == "LV"` finds the Raiders in their Oakland years too, and "LA" the Rams in their St. Louis years.

Schedules (`load_schedules()`) have one row per game, keyed by `game_id`, with the final score, the betting line, the stadium, the roof, the weather, the starting quarterbacks, the head coaches and even the referee. They are how you attach game-level facts (was it a divisional game? on short rest?) to plays.

Rosters (`load_rosters()`) have one row per player per team per season, keyed by `gsis_id`, the league's player ID (the 00-00... strings). Play-by-play stores the same ID in two sets of player columns, so a merge gives you a player's position, size, college or draft slot. The `passer_player_id`, `rusher_player_id` and `receiver_player_id` columns follow the box score: a scrambling quarterback is the `rusher_player_id`. The shorter `passer_id`, `rusher_id` and `receiver_id` follow the `pass` and `rush` flags: a scrambling quarterback is the `passer_id`, and `rusher_id` is empty. Use the first set to count a player's carries the way the box score does, the second to credit dropbacks to the quarterback.

```
sched = nflr.load_schedules([2025]).to_pandas()
sb = sched[sched.game_id == "2025_22_SEA_NE"]
sb_cols = ["game_id", "gameday", "away_team", "away_score", "home_team", "home_score",
           "spread_line", "roof", "away_coach", "home_coach", "referee"]
display(md_table(sb[sb_cols].T.reset_index().set_axis(["column", "value"], axis=1).astype(str)))

roster = nflr.load_rosters([2025]).to_pandas()
carriers = (df25[(df25.game_id == "2025_22_SEA_NE") & (df25.play_type == "run") & (df25.rush == 1)]
            .merge(roster[["gsis_id", "position", "college"]].drop_duplicates("gsis_id"),
                  left_on="rusher_player_id", right_on="gsis_id", how="left")
            .groupby(["posteam", "rusher", "position", "college"])
            .agg(carries=("play_id", "size"), yards=("yards_gained", "sum"),
```

```
epa=("epa", "sum"))
.reset_index().sort_values("carries", ascending=False).head(5))
display(md_table(carriers))
```

column	value
game_id	2025_22_SEA_NE
gameday	2026-02-08
away_team	SEA
away_score	29
home_team	NE
home_score	13
spread_line	-4.5
roof	outdoors
away_coach	Mike Macdonald
home_coach	Mike Vrabel
referee	Shawn Smith

posteam	rusher	position	college	carries	yards	epa
SEA	K.Walker	RB	Michigan State; Wake Forest	27	135	-2.04
NE	R.Stevenson	RB	Oklahoma; Cerritos College	7	23	-1.60
NE	T.Henderson	RB	Ohio State	6	19	-1.31
SEA	G.Holani	RB	Boise State	2	6	0.17
SEA	R.Shaheed	WR	Weber State	1	-5	-1.28

(Seattle was the designated road team, and `spread_line` is in the home team's favour, so -4.5 means the "home" Patriots were 4.5-point underdogs.) Walker's line from the merge matches the box score (27 carries, 135 yards), which is the first check you should always make after a join: does a total you already know come out right? It also holds a small surprise. The game's MVP totals slightly *negative* EPA on his carries. Only 26% of them were successes (positive EPA), his median carry gained 2 yards, and 15 of his 27 gained two yards or fewer; each of those cost a little, and the 30- and 29-yard bursts did not quite pay for them. That is the gap between yards and value that Expected Points, EPA, and Success Rate is about.

Participation data (`load_participation()`) is the table that powers the personnel and formation charts in Part 2. It has one row per play, keyed by `nflverse_game_id` and `play_id`, and lists the offensive and defensive personnel, the formation, the number of defenders in the box, the number of pass rushers, the players' IDs and, from 2023, their names, positions and jersey numbers. For most dropbacks from 2018 on it also carries the coverage: `defense_man_zone_type` (man or zone) and `defense_coverage_type` (COVER_1, COVER_3 and so on). Everything you do with it is shaped by where it comes from: two suppliers, Next Gen Stats for 2016–2022 and FTN from 2023, and it is published **only after a season ends**: FTN releases each season's file under a Creative Commons licence (CC-BY-SA 4.0) once the season is over, so the 2026 file will not exist until about February 2027.⁸

FTN charting data (`load_ftn_charting()`) is the other half of FTN's work: a human charter's notes on every play, from 2022 on, updated **weekly during the season**. It records where the quarterback lined up (`qb_location`: U for under center, S for shotgun, P for pistol), how many players were in the backfield and in the box, whether there was motion, play-action, a screen or an RPO, how many defenders blitzed, and judgement calls on the throw (catchable? contested? dropped? interception-worthy?). It joins on

⁸nflverse, participation data dictionary: https://nflreadr.nflverse.com/articles/dictionary_participation.html; release notes for the FTN-sourced files (2023 and 2024 re-released in September 2025, 2025 in February 2026) at <https://github.com/nflverse/nflverse-data/releases>; per the course fact sheet, checked October 2026.

`nflverse_game_id` and `nflverse_play_id`. It does not have the coverage; that lives only in participation.⁹

Watch two of its definitions. `is_motion` means motion **before or at** the snap, not only motion at the snap, so it is not the same thing as the “motion at the snap” rates you’ll hear on broadcasts, nor the same as PFF’s shift-and-motion rate. And there is no flag for the tush push: use `is_qb_sneak`, which covers both the classic sneak and the pushed version. FTN does not publish how often two of its charters agree on the judgement columns (catchable, contested, interception-worthy), so treat them as one trained viewer’s opinion: good for season totals, shaky for any single play.

Put the two timetables together and you get a gap that matters for the Sunday notebook: **during the season, nflverse has no personnel groupings**. Participation for 2026 arrives after the season; FTN charting arrives weekly but records the backfield count (`n_offense_backfield`), the quarterback’s spot and the box count, not how many backs and tight ends were on the field. If you want this week’s personnel, you log it yourself from the broadcast, which is exactly what the prediction log is for.

Joining participation, and the 2022-to-2023 source break

Joining is one merge. The care is in the checking. Here are ten seasons of runs and dropbacks joined to participation, with the match rate by season:

```
SEASONS = list(range(2016, 2026))
part = nflr.load_participation(SEASONS).to_pandas()
plays = snaps[snaps.season.isin(SEASONS)]
joined = plays.merge(part, left_on=["game_id", "play_id"],
                    right_on=["nflverse_game_id", "play_id"], how="left", indicator=True)
check = joined.groupby("season").agg(
    plays=("play_id", "size"),
    matched=("_merge", lambda s: (s == "both").mean()),
    personnel=("offense_personnel", lambda s: s.notna().mean()),
    formation=("offense_formation", lambda s: s.notna().mean()),
    coverage=("defense_coverage_type", lambda s: s.notna().mean()))
md_table(check.reset_index(), digits=3)
```

season	plays	matched	personnel	formation	coverage
2016	32,291	1.000	1.000	0.990	0.000
2017	31,987	1.000	1.000	0.992	0.000
2018	31,751	1.000	1.000	0.991	0.551
2019	32,047	1.000	1.000	0.999	0.553
2020	32,444	1.000	1.000	0.996	0.551
2021	33,986	1.000	1.000	0.993	0.547
2022	33,652	1.000	1.000	0.999	0.534
2023	33,836	1.000	1.000	0.997	0.610
2024	33,335	1.000	1.000	0.999	0.603
2025	32,813	1.000	1.000	0.998	0.601

Every run and dropback finds its participation row, and almost all of them have personnel and a formation. Coverage is missing for 2016–2017 entirely and present on a little over half of runs and dropbacks after that, because it is charted for dropbacks only. So far, so good. Now look at *what* the formation column says, season by season:

⁹nflverse, FTN charting data dictionary: https://nflreadr.nflverse.com/articles/dictionary_ftn_charting.html (`is_motion`: “motion occurred on the play before or at the time of the snap”). FTN Data: <https://ftndata.com/>. Licences: the nflreadpy README says most nflverse data is CC-BY 4.0 and the FTN data CC-BY-SA 4.0 (<https://github.com/nflverse/nflreadpy#license>); nflreadr 1.5.0’s release notes say the FTN-supplied participation data is CC-BY-SA 4.0 and should be credited “to FTN Data via nflverse” (<https://cran.r-project.org/web/packages/nflreadr/news/news.html>). Credit FTN charting the same way.

```

form = joined[joined.offense_formation.notna()].copy()
form["label"] = form.offense_formation.replace({"JUMBO": "JUMBO / WILDCAT",
                                                "WILDCAT": "JUMBO / WILDCAT"})

ORDER = ["SHOTGUN", "EMPTY", "PISTOL", "SINGLEBACK", "I_FORM", "JUMBO / WILDCAT", "UNDER CENTER"]
fshare = pd.crosstab(form.season, form.label, normalize="index").reindex(columns=ORDER,
                                                                    fill_value=0) * 100

plot_formation(fshare)
plt.show()

```

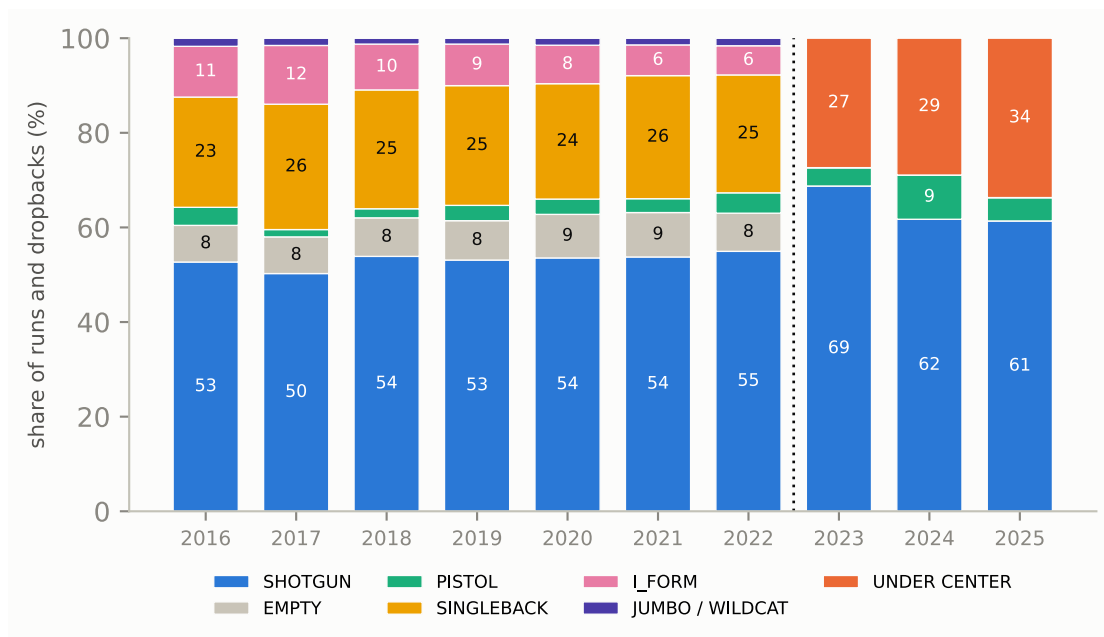


Figure 7: The offense_formation labels in nflverse participation data, share of regular-season runs and dropbacks by season. Through 2022 (Next Gen Stats) the labels describe the backfield: shotgun, singleback, I-formation, empty, pistol, jumbo, wildcat. From 2023 (FTN) there are only three, and they describe where the quarterback stood: shotgun, under center, pistol. ‘Shotgun’ jumps from 55% to 69% across the dotted line, mostly because empty backfields (grey, 8% in 2022) were folded into it; the remaining few points are a real rise in shotgun that the play-by-play flag also shows. The football did not change that much in one off-season; the labels did.

That chart is a **source break**: a point in a series where the supplier or the method of recording changes, so the two sides are not directly comparable. Nothing in the football explains a formation vocabulary shrinking from seven words to three over one off-season. Next Gen Stats described the backfield; FTN describes the quarterback’s spot. An I-formation and a singleback set are both “UNDER CENTER” now, and an empty backfield, in which the quarterback is nearly always in the shotgun, is now “SHOTGUN.”

The break shows up in other columns too, and each one has its own trap:¹⁰

- **Personnel strings change format.** Through 2022 they read "1 RB, 2 TE, 2 WR", listing linemen only when there are more than five. From 2023 they list every player by roster position, "1 C, 2 G, 1 QB, 1 RB, 2 T, 2 TE, 2 WR", so any code that splits the string has to handle both, as the reproduction below does.
- **Coverage labels change.** FTN added Cover 9 (a split-field coverage), COMBO (a call that mixes man and zone; see combo coverage) and BLOWN (a coverage the charter judged had broken down), and dropped PREVENT (the soft prevent defense of the last minutes).
- **Some columns simply stop.** `ngs_air_yards` is empty from 2024 on; use play-by-play’s own `air_yards` instead, which has no break.
- **The man-zone split jumps** in a way that looks like coding, not football:

¹⁰Course calculation from nflverse participation data, regular-season runs and dropbacks 2016-2025 (formation labels, personnel strings, coverage labels and man share by season), matching the course fact sheet’s checks of October 2026.

```
mz = joined[(joined["pass"] == 1) & joined.defense_man_zone_type.isin(["MAN_COVERAGE",
                                                                       "ZONE_COVERAGE"])]
man = mz.groupby("season").defense_man_zone_type.apply(lambda s: (s == "MAN_COVERAGE").mean())
print((man * 100).round(1).to_string())
```

```
season
2018    35.0
2019    37.2
2020    34.4
2021    30.3
2022    28.8
2023    41.2
2024    48.3
2025    30.8
```

Man coverage on 29% of labelled dropbacks in 2022, then 41% and 48% under the new supplier, then back to 31% in 2025. Defenses did not double their man coverage and then abandon it; the charters changed how they called borderline coverages. A chart that ran straight across 2022–2025 would invent a trend. (Tendencies: PROE, Personnel, Motion, and Coverage builds coverage charts with this caveat built in.)

Three rules for a source break

1. **Mark it.** Any chart that crosses 2022–2023 in participation data gets a dotted line and a note, as the personnel chart below does. A reader can then judge the jump for themselves.
2. **Translate to what both sides can say.** When the labels disagree, drop to the coarser category both suppliers can express. Next Gen Stats never said “under center”, but three of its labels mean it: SINGLEBACK, I_FORM and JUMBO all put the quarterback under the center. Collapse them, and the 2016–2022 labels speak FTN’s language.
3. **Cross-check with an independent source.** Play-by-play has its own `shotgun` flag, parsed from the statisticians’ play description, and FTN’s `qb_location` is a separate charter’s call. If they agree with each other across the years, the trend is real.

Here are rules 2 and 3 at work on the “under-center comeback” that this book describes in The Modern Era. “participation (U)” is the share of snaps whose formation label means under center, translated as in rule 2; “pbp (not gun)” is the share the statisticians did *not* mark “(Shotgun)”; “FTN (U)” and “FTN (P)” are FTN charting’s under-center and pistol shares. The play-by-play and FTN charting columns come from independent recorders:

```
UNDER = {"SINGLEBACK", "I_FORM", "JUMBO", "UNDER CENTER"} # labels meaning "QB under center"
ftn = nflr.load_ftn_charting([2022, 2023, 2024, 2025]).to_pandas()
ftn["qb_location"] = ftn.qb_location.astype(str).str.strip()
fj = plays[plays.season >= 2022].merge(
    ftn[["nflverse_game_id", "nflverse_play_id", "qb_location", "is_motion", "is_rpo"]],
    left_on=["game_id", "play_id"], right_on=["nflverse_game_id", "nflverse_play_id"],
    how="left")
xc = pd.DataFrame({
    "participation (U)": form.groupby("season").offense_formation.apply(
        lambda s: s.isin(UNDER).mean()),
    "pbp (not gun)": 1 - plays.groupby("season").shotgun.mean(),
    "FTN (U)": fj.groupby("season").qb_location.apply(lambda s: (s == "U").mean()),
    "FTN (P)": fj.groupby("season").qb_location.apply(lambda s: (s == "P").mean())})
md_table(xc.map(lambda v: f"{v:.1%}" if pd.notna(v) else "-").reset_index())
```

season	participation (U)	pbp (not gun)	FTN (U)	FTN (P)
2016	35.3%	35.8%	–	–
2017	40.1%	40.7%	–	–

season	participation (U)	pbp (not gun)	FTN (U)	FTN (P)
2018	35.8%	36.1%	–	–
2019	35.1%	35.3%	–	–
2020	33.7%	33.9%	–	–
2021	33.5%	33.8%	–	–
2022	32.0%	32.1%	31.6%	3.1%
2023	27.4%	27.8%	27.3%	3.9%
2024	28.9%	29.4%	28.8%	9.3%
2025	33.7%	33.8%	33.6%	4.9%

Read across a row and the three columns agree to within a point or two in every season. Translated, the formation label no longer jumps at the break: under center falls from about 32% in 2022 to 27% in 2023, exactly as the play-by-play flag says (shotgun peaked at about 72% in 2023), and then climbs back to about 34% in 2025 in all three columns. 2022 is the best test, because both suppliers charted it: Next Gen Stats, translated, says 32.0% and FTN says 31.7%. Three recorders, one trend: quarterbacks really did move back under center.¹¹ (The play-by-play flag counts the pistol as shotgun, which is why “not shotgun” lines up with “under center”).

The last column is a warning of a different kind. FTN’s pistol share goes from 4% to 9% and back to 5% in consecutive seasons, with no change of supplier. A league-wide formation does not double and halve in two years; that is the shape of charters changing how they call a borderline alignment. A series can break without a new supplier. Even the play-by-play flag, which has kept one supplier since 1999, is typed by a different home stadium’s crew each week, so a careful analyst treats it as about thirty recorders who mostly agree.

They do not always agree on single plays, though, and Walker’s run is one of the exceptions:

```
wf = ftn[(ftn.nflverse_game_id == "2025_22_SEA_NE") & (ftn.nflverse_play_id == 1015)].iloc[0]
wp_ = part[(part.nflverse_game_id == "2025_22_SEA_NE") & (part.play_id == 1015)].iloc[0]
print(f"pbp description:           {walker.desc[:30]}...   shotgun flag = {walker.shotgun:.0f}")
print(f"participation formation: {wp_.offense_formation} personnel: {wp_.offense_personnel}")
print(f"FTN charting:               qb_location = {wf.qb_location}, is_rpo = {wf.is_rpo}, "
      f"defenders in box (participation) = {wp_.defenders_in_box:.0f}")
both = fj[fj.season == 2025]
n_dis = int(((both.shotgun == 1) & (both.qb_location == "U")).sum())
print(f"2025 runs and dropbacks where pbp says shotgun but FTN says under center: {n_dis}")
```

```
pbp description:           (14:04) (Shotgun) 9-K.Walker l...   shotgun flag = 1
participation formation: UNDER CENTER personnel: 1 C, 2 G, 1 QB, 1 RB, 2 T, 1 TE, 3 WR
FTN charting:               qb_location = U, is_rpo = True, defenders in box (participation) = 8
2025 runs and dropbacks where pbp says shotgun but FTN says under center: 43
```

The two records are made differently. The statisticians type each play live from the press box, as the game is played; FTN’s charters work from video later and can replay the snap. The disagreements cluster on alignments that sit between the categories: a quarterback in the pistol, a few yards behind the center, or one who starts under center and steps back before the snap. Here the statisticians typed “(Shotgun)” and FTN’s charter marked the quarterback under center and called the play a run-pass option. Without the film you can’t say who is right, but you can say why they differ. It happens on a few dozen plays a season, too few to move a trend, and it is a useful reminder: every column in these tables is somebody’s judgement, recorded at speed.

¹¹Course calculation from nflverse play-by-play (shotgun) and FTN charting (qb_location), regular-season runs and dropbacks with a down, 2016-2025. The same trend in other sources: NFL Next Gen Stats reported offenses using fewer than three wide receivers on 41.7% of 2025 plays (James Reber, NFL.com, “NFL trend watch: heavy personnel could shape Super Bowl LX showdown between Patriots and Seahawks”): <https://www.nfl.com/news/nfl-trend-watch-heavy-personnel-could-shape-super-bowl-lx-showdown-between-patriots-and-seahawks>.

Reproducing a chart end to end: personnel by season

Now put it together. Personnel showed how the league's mix of personnel groupings moved from 2016 to 2025. Here is that chart rebuilt from nothing but the cells above: the `plays` funnel, the participation join, and a parser that reads both personnel formats.

```
def count(s, positions):
    """Sum the counts for some positions in a personnel string, in either format:
    2016-2022 (NGS) '1 RB, 2 TE, 2 WR'; 2023-2025 (FTN) '1 C, 2 G, 1 QB, 1 RB, 2 T, 2 TE, 2 WR'."""
    if not isinstance(s, str):
        return np.nan
    return sum(int(n) for n, pos in re.findall(r"(\d+)\s+([A-Z]+)", s) if pos in positions)

pers = joined[["season", "offense_personnel"]].copy()
pers["rb"] = pers.offense_personnel.map(lambda s: count(s, {"RB", "FB"})) # fullbacks are backs
pers["te"] = pers.offense_personnel.map(lambda s: count(s, {"TE"}))
pers["ol"] = pers.offense_personnel.map(lambda s: count(s, {"OL", "C", "G", "T"}))
pers = pers.dropna(subset=["rb", "te"])
pers["pers"] = pers.rb.astype(int).astype(str) + pers.te.astype(int).astype(str)
share = pd.crosstab(pers.season, pers.pers, normalize="index") * 100
share["6+ OL"] = pers.groupby("season").ol.apply(lambda n: (n > 5).mean() * 100)
md_table(share[["11", "12", "13", "21", "6+ OL"]].round(1).reset_index(), digits=1)
```

season	11	12	13	21	6+ OL
2016	60.7	18.8	3.1	7.9	5.7
2017	59.6	20.6	4.2	8.4	3.6
2018	64.6	17.8	3.5	8.0	3.2
2019	60.3	20.9	3.0	7.9	4.3
2020	59.4	20.1	3.6	8.3	2.6
2021	60.5	21.6	3.6	7.3	3.9
2022	62.8	20.1	4.1	8.2	3.1
2023	64.2	20.4	3.4	7.4	2.6
2024	62.7	23.5	3.7	6.3	2.9
2025	58.8	24.4	5.1	7.0	5.4

```
plot_personnel(share, {"11": style.OFFENSE, "12": style.DEFENSE, "21": style.THIRD, "13": "#eda100"})
plt.show()
```

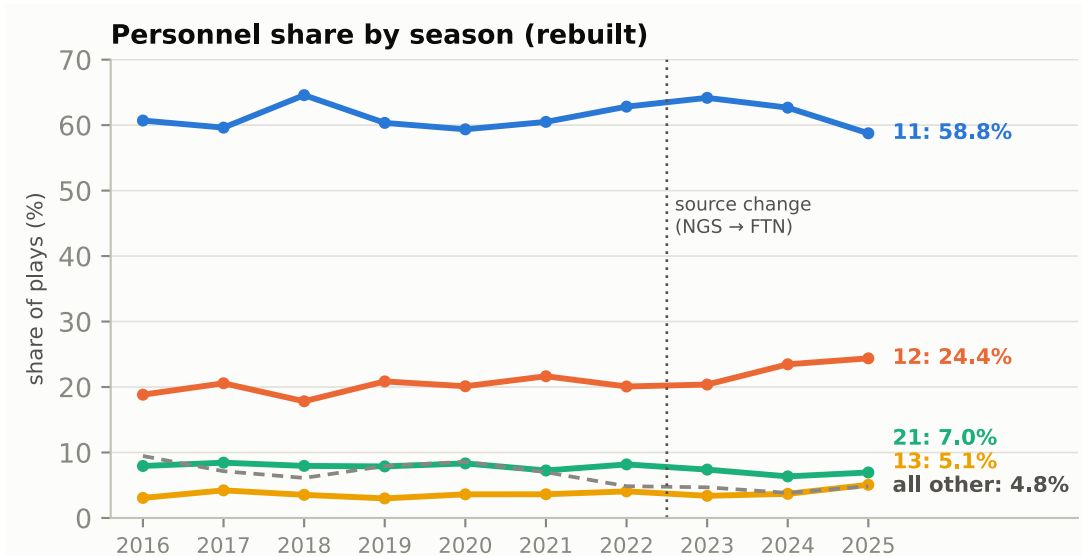


Figure 8: The personnel chart from 02-01, rebuilt end to end in this chapter: share of NFL regular-season runs and dropbacks by offensive personnel, 2016–2025, all of them, with no garbage-time filter, as in 02-01 (nflverse play-by-play joined to participation data; backs = RB + FB, then tight ends). 11 personnel held near 60% for a decade and fell to 58.8% in 2025; 12 personnel rose to 24.4%. The dotted line marks the change of supplier; personnel survives it, because both suppliers count backs and tight ends.

Same numbers as 02-01, to the decimal: 58.8% for 11 personnel in 2025, 24.4% for 12, 5.1% for 13. That is the point of the exercise. A chart you can rebuild is a chart you can extend: change `SEASONS`, filter `joined` to one team (`joined.posteam == "SEA"`), or split by down, and the same dozen lines answer a new question.

Personnel strings record **who** was on the field by roster position, not **what they did**, and a coach would add three limits. A sixth offensive lineman reporting as an eligible player (jumbo) is invisible to the backs-and-tight-ends count, so a six-lineman short-yardage set is filed as 11 or 12 personnel; the last column shows that happens on 3–6% of snaps. A player listed as a fullback counts as a back even when he lines up as a tight end, and a tight end used as a fullback counts as a tight end, so the label follows the roster, not the alignment. And personnel is not formation: 12 personnel with one tight end split wide looks to the defense much like 11. The count survives the source break; the meaning was never in the count.

When you reproduce someone else’s chart, three things usually explain a mismatch, in this order: **the filter** (regular season only? penalties? scrambles?), **the counting rule** (here, does a fullback count as a back? the course says yes), and **the data vintage** (nflverse occasionally re-releases past seasons; FTN re-released 2023 and 2024 participation in September 2025). Check them before you suspect the arithmetic.

Film room: league pass rate by down, reproduced

The first chart in this book that predicted anything was Downs, Distance, and Field Position’s grid of how often offenses passed by down and distance in 2025. Here it is again, rebuilt with that chapter’s filter (win probability 20% to 80%, outside the last two minutes of each half) and its buckets:

```
n25 = snaps[(snaps.season == 2025) & (snaps.half_seconds_remaining > 120)
            & snaps.wp.between(0.2, 0.8) & (snaps.down <= 3)].copy()
BUCKETS = ["1", "2–3", "4–6", "7–10", "11+"]
n25["dist"] = pd.cut(n25.ydstogo, [0, 1, 3, 6, 10, 99], labels=BUCKETS)
grid = n25.groupby(["down", "dist"], observed=False)["pass"].agg(["mean", "size"]).reset_index()
plot_pass_grid(grid, BUCKETS)
plt.show()
```

	1 yd	2–3 yds	4–6 yds	7–10 yds	11+ yds
1st down	rare	rare	rare	46% 7,445 plays	69% 318 plays
2nd down	26% 344 plays	32% 645 plays	48% 1,543 plays	69% 2,710 plays	82% 891 plays
3rd down	21% 487 plays	64% 678 plays	93% 914 plays	97% 1,072 plays	92% 681 plays

Figure 9: 01-02’s pass-rate grid, rebuilt: how often the offense passed, by down and distance, 2025 regular season, with 01-02’s filter (the 20–80% neutral band, outside the last two minutes of each half). A pass is any dropback, so sacks and scrambles count. Cells with fewer than 250 plays are marked ‘rare’. Source: nflverse play-by-play.

Same grid, same numbers. Now do what 01-02 couldn’t, because it had no code: ask whether the pattern is new. Play-by-play goes back to 1999, so the same question can be asked of 27 seasons. This is a league-wide history, so the cell uses the default garbage-time filter (the two filters differ by a point or two, which a 27-season trend line can absorb). It tracks the first three downs; fourth down mixes “how often do they pass?” with “how often do they go for it at all?”, which is Win Probability and Decision Analysis’s subject:

```
gt = snaps[snaps.wp.between(0.10, 0.90) & (snaps.half_seconds_remaining > 120)]
by = gt.groupby(["season", "down"])[["pass"]].mean().unstack() * 100
plot_history(by)
plt.show()
print(by.loc[[1999, 2009, 2019, 2025], [1, 2, 3, 4]].round(1).to_string())
```

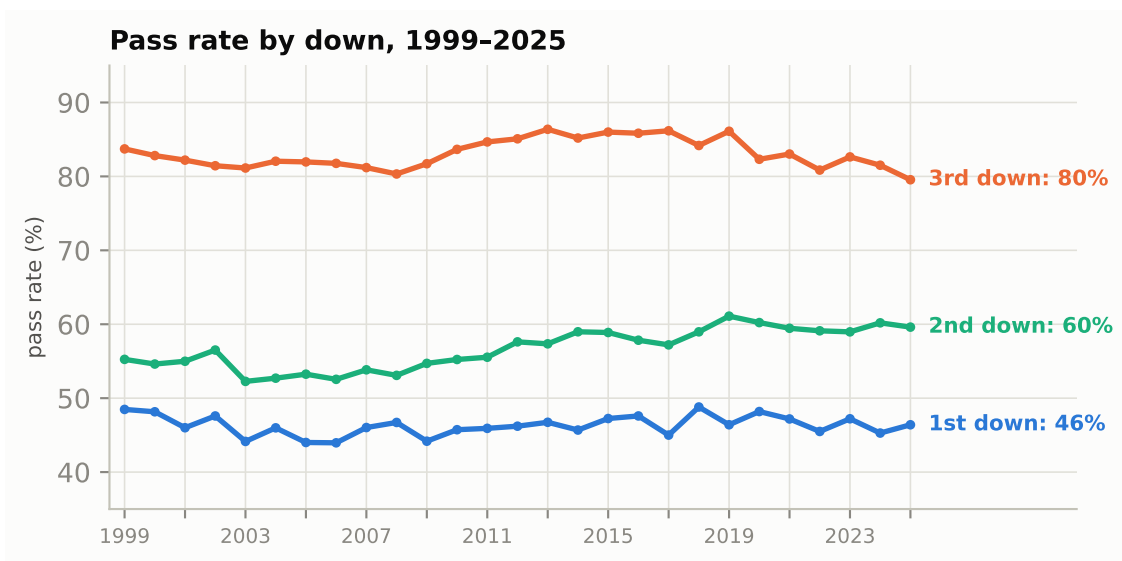


Figure 10: League pass rate by down, every regular season from 1999 to 2025, with the course’s garbage-time filter (win probability 10–90%, outside the last two minutes of each half); a pass is any dropback, including sacks and scrambles. First down has hardly moved in 27 seasons. Second down drifted up by about five points. Third down peaked around 2013–2019 and has eased since. Source: nflverse play-by-play.

down	1.0	2.0	3.0	4.0
season				
1999	48.5	55.2	83.7	42.7

2009	44.2	54.7	81.7	39.7
2019	46.4	61.1	86.1	54.5
2025	46.4	59.6	79.5	56.8

A falling third-down pass rate could mean two different things: more third downs that are short (so the same coaches run more), or the same distances with a different call. Split the third downs by distance and the two stories separate:

```
t3 = gt[(gt.down == 3) & (gt.season.between(2013, 2019) | (gt.season == 2025))].copy()
t3["era"] = np.where(t3.season == 2025, "2025", "2013-19")
t3["dist"] = pd.cut(t3.ydstogo, [0, 1, 3, 6, 10, 99], labels=BUCKETS)
mix = pd.crosstab(t3.dist, t3.era, normalize="columns") * 100 # share of 3rd downs
call = t3.groupby(["dist", "era"], observed=True)["pass"].mean().unstack() * 100 # pass rate
split = pd.concat({"share of 3rd downs (%)": mix, "pass rate (%)": call}, axis=1).round(1)
split.columns = [f"{a}, {b}" for a, b in split.columns]
md_table(split.reset_index(names="yards to go"), digits=1)
```

yards to go	share of 3rd downs (%), 2013-19	share of 3rd downs (%), 2025	pass rate (%), 2013-19	pass rate (%), 2025
1	11.0	13.0	27.2	21.3
2-3	15.9	17.6	80.2	64.2
4-6	24.7	24.2	95.8	93.1
7-10	29.4	27.2	97.9	97.2
11+	18.9	18.1	92.3	91.5

i Film room: the league, 1999-2025

What to notice in charts you built yourself:

- **First down is the most stable number in football, on average.** In every season from 1999 to 2025, with the game still in doubt, offenses passed on between 44% and 49% of first downs. Twenty-seven seasons of rule changes that favoured the passing game (the illegal-contact emphasis of 2004, the protection of defenseless receivers and quarterbacks) and the spread of shotgun offenses barely moved it. That is an average over every team, personnel group and formation, not a coin flip on every snap: the scorebug alone tells you almost nothing on first-and-10, but the formation tells you a lot, which is why Part 2 exists.
- **The passing boom showed up on second down.** Second-down pass rate climbed from about 55% to about 60%. Coaches who once ran on second down to “set up third-and-manageable” increasingly throw to convert on second down instead.
- **Third down eased after the late 2010s, and it was the call, not the distance.** Third-down pass rates peaked around 86% in 2013-2019 and fell below 80% in 2025. The split table shows the mix of third-down distances barely moved; what changed is the call on short yardage, above all third-and-2 or 3, where the pass rate fell from about 80% to about 64%. That fits more under-center play and more quarterback sneaks and pushes; the chapters on third down and the modern era argue the causes.
- **Fourth down rose most of all,** from about 43% passes in 1999 to about 57% in 2025, and not because teams go for it at longer distances (the typical fourth-down snap is, if anything, shorter). Offenses now throw on about nine in ten fourth downs of two yards or more, where 1999 offenses often ran.¹²

The lesson for reading a game is the one the 01-02 grid taught: from the scorebug alone, first-and-10 is close to a coin flip and third-and-long is not, and the formation is what breaks the coin flip (How to Watch a Broadcast measured how much). The lesson for working with data is new: the 2025 grid was one season’s snapshot, a few lines of code turned it into a 27-season history, and one more table told two explanations apart. That is where most good questions start.

¹²Course calculation from nflverse play-by-play, regular-season fourth-down runs and dropbacks with the course’s garbage-time filter: 171 snaps in 1999 (30% at four or more yards to go; pass rate 55% at 2 yards, 67% at 3, 77% at 4 or more) and 437 in 2025 (18% at four or more; 89%, 96% and 95%).

From a row to a diagram: nflverse meets gridiron

The book's diagram library, `gridiron`, and `nflverse` speak almost the same language. A row gives you the situation (`yardline_100`, `ydstogo`), and participation and FTN add the personnel, the formation and the box count. What no row gives you is **where each player stood**: that needs tracking data, the subject of Tracking Data and the Big Data Bowl. So a diagram built from a row is part data and part illustration, and the caption has to say which part is which. Here is Walker's run, built from its row:

```
los = 110 - float(walker.yardline_100)
off = offense("singleback") # 11 personnel, QB under center (FTN: U)
dfn = defense("nickel", "single_high", off) # 3 CB, 2 safeties: five DBs
# gridiron's default puts the nickel over the tight end; a real nickel apexes the slot (H)
dfn = [d.moved(d=4.5, w=-7.0) if d.key == "NB" else d for d in dfn]
grouping = (f"{count(wp_.offense_personnel, {'RB', 'FB'})}"
            f"{count(wp_.offense_personnel, {'TE'})} personnel")

def walker_play(pursuit=False):
    """The row as a Play. pursuit=True adds a typical QB fake and defensive pursuit (animation)."""
    p = Play(off, dfn, los=los, to_go=float(walker.ydstogo),
            title=f"[', '1st', '2nd', '3rd', '4th'][int(walker.down)] & {walker.ydstogo:.0f} "
            f"at SEA {110 - walker.yardline_100 - 10:.0f} · {grouping} (from the data)")
    p.handoff("RB", t=0.8)
    p.run("RB", [(2.5, -3.0), (7.0, -7.8), (12.0, -10.5), (31.0, -13.0)]) # off the top: +30
    p.block("LT", target="WDE") # hook the end
    p.block("H", target="NB") # slot seals the nickel inside
    p.block("X", target="LCB") # stalk the corner
    p.highlight("RB")
    if pursuit:
        p.path("QB", [(-1.2, 0.8), (-1.5, 4.8)], delay=0.9) # carry-out fake
        p.path("WILL", [(1.5, -4.0), (5.5, -7.0)], delay=0.5) # fast flow, over the top
        p.path("MIKE", [(1.5, -3.5), (5.0, -6.5)], delay=0.6)
        p.path("LCB", [(0.6, 0.3)], delay=0.9) # fights the stalk block
        p.path("FS", [(1.0, -5.0), (2.5, -10.0)], delay=0.6) # alley, then the angle
        p.path("SS", [(1.0, -5.0)], delay=0.7) # backside pursuit
    return p

row_play = walker_play()
row_play.note(f"+{walker.yards_gained:.0f} yds ↑", at=(12.5, -6.5))
fld = row_play.draw(window=(-9, 17), lateral=(-20, 20))
drop_numbers(fld.ax)
corners = [row_play.to_abs(d_, w_) for d_, w_ in [(0.1, -8.4), (8.2, -8.4), (8.2, 7.6), (0.1, 7.6)]]
bx, by_ = fld.xy(np.array([c[0] for c in corners]), np.array([c[1] for c in corners]))
fld.ax.add_patch(plt.Polygon(np.c_[bx, by_], closed=True, fill=False, ec=style.DEFENSE_DARK,
                             lw=1.0, ls=(0, (3, 2)), zorder=3))
tx, ty = fld.xy(*row_play.to_abs(3.0, 8.4))
fld.ax.text(tx, ty, "← the box: 8 defenders", fontsize=7, color=style.DEFENSE_DARK, ha="left",
            va="center", zorder=7, bbox=dict(boxstyle="round,pad=0.2", fc=style.SURFACE,
            ec="none"))
gx, gy = fld.xy(*row_play.to_abs(-5.0, 0.0))
fld.ax.add_patch(Circle((gx, gy), 0.9, fill=False, ec=style.OFFENSE_DARK, lw=1.0, ls=(0, (2, 2)),
                        zorder=7))
fld.ax.text(gx + 1.6, gy, "pbp: '(Shotgun)'", fontsize=6.8, color=style.OFFENSE_DARK,
            va="center", zorder=7)
plt.show()
```

2nd & 10 at SEA 24 · 11 personnel (from the data)

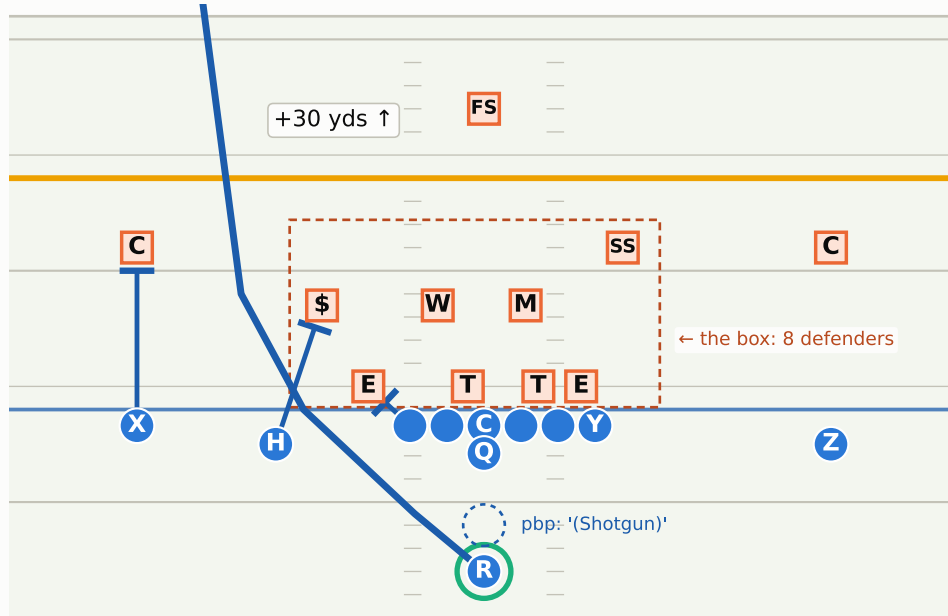


Figure 11: Walker's run drawn from its data. From the row: second-and-10, ball at yardline₁₀₀ = 76 (so the line of scrimmage is at BDB $x = 34$ and the yellow line 10 yards on), a run to the left end for 30 yards, to the New England 46; the arrow runs off the top of the diagram. From participation: 11 personnel (one back, one tight end, three receivers) against a nickel defense, five defensive backs (three corners, two safeties), with eight defenders in the box (dashed outline): the nickel back (\$) apexed over the slot and the strong safety down. Participation lists the edge players as outside linebackers; they are labelled E. From FTN: quarterback under center, and a run-pass option (the throw it offered isn't recorded, so none is drawn). Everything else (each player's exact spot, the blocking) is the course's typical version, not the real one. The dashed ring is where the play description's '(Shotgun)' would put the quarterback.

The title is built from the data, so the same cell draws any row you point it at. The picture also produces **simulated tracking data**: positions ten times a second, in exactly the column layout the Big Data Bowl uses. That is how this book animates hand-drawn plays, and it means the code that animates them will animate real tracking frames unchanged. Here are a few of the running back's frames from the animated version, which adds a typical quarterback fake and defensive pursuit:

```
anim_play = walker_play(pursuit=True)
trk = anim_play.to_tracking(game_id=2026020800, play_id=1015) # simulated, BDB-format
print(f"{len(trk):,} rows: {trk.frameId.nunique()} frames x {trk.displayName.nunique()} "
      f"tracks (22 players + the ball)")
rb = trk[trk.displayName == "RB"].iloc[[0, 5, 8, 12, 20, 36]]
md_table(pd.DataFrame({"frameId": rb.frameId, "frameType": rb.frameType,
                       "time": rb.time.map(lambda v: f"{v + 0.0:.1f}"),
                       "x": rb.x.round(1), "y": rb.y.round(1), "s": rb.s.round(1),
                       "dir": rb.dir.map(lambda v: "-" if pd.isna(v) else f"{v:.0f}"),
                       "event": rb.event.fillna("")})), digits=1)
```

1,518 rows: 66 frames x 23 tracks (22 players + the ball)

frameId	frameType	time	x	y	s	dir	event
1	BEFORE_SNAP	-0.5	27.0	26.7	0.0	-	
6	SNAP	+0.0	27.0	26.7	3.2	40	ball_snap
9	AFTER_SNAP	+0.3	28.2	28.2	6.5	40	

frameId	frameType	time	x	y	s	dir	event
13	AFTER_SNAP	+0.7	29.9	30.1	6.5	43	
21	AFTER_SNAP	+1.5	33.5	33.9	6.5	43	
37	AFTER_SNAP	+3.1	42.9	37.7	6.5	83	

Each row is one player at one moment. `frameType` says whether the frame is before, at or after the snap; `time` counts seconds from the snap. `x` runs downfield (110 is the goal line Seattle is attacking) and `y` runs across the field from 0 to 53.3, increasing toward the offense's left, which is why a run to the left end makes `y` climb. `s` is speed in yards per second, and `dir` is the direction of travel in degrees, clockwise from the +`y` direction: 0 is toward the offense's left sideline, 90 straight downfield, 180 toward the right sideline.

```
trk_show = trk.copy()
ball_on = (trk_show.side == "ball") & (trk_show.time > 0)
trk_show.loc[ball_on, "y"] += 1.1 * np.clip(trk_show.loc[ball_on, "time"] / 0.15, 0, 1) # tucked, beside
the marker
show_animation_clean(trk_show, times=[0.0, 0.8, 2.2, 3.6], title="Simulated, BDB-format",
                    to_go=float(walker.ydstogo))
```

Simulated, BDB-format

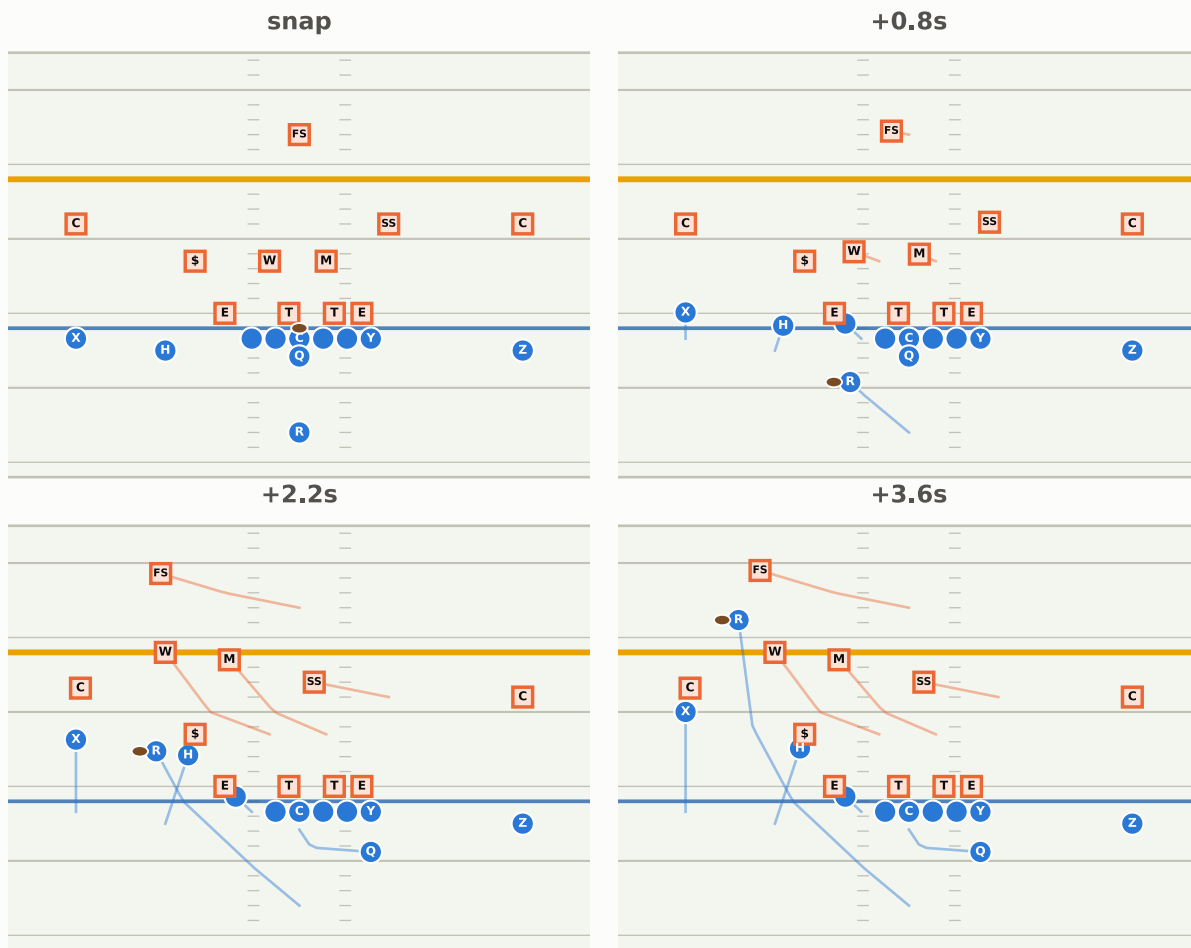


Figure 12: Walker's run as simulated, BDB-format tracking (generated from the diagram above, not real player positions). Only the ball carrier's path comes from the data; the blocks, the quarterback's fake and the defenders' pursuit are a typical version. Snap; the handoff at +0.8 s (the football is drawn beside the carrier so his R stays visible); at +2.2 s the back has bounced outside the hooked end and the slot's block on the nickel while the linebackers flow and the free safety comes down; by +3.6 s he is past the yellow line with the Will and the free safety closing. In the web version this is an animation; in print, a frame strip.

With real Big Data Bowl files the drawing code is the same; only the source of the frames changes. The cell below is not run in this book (the Kaggle files are not redistributed), but it is what you will write in 13-05:

```
import pandas as pd
from gridiron import bdb, show_animation

trk = bdb.load_tracking("data/bdb/tracking_week_1.csv")    # BDB 2025: the 2022 season
plays_bdb = pd.read_csv("data/bdb/plays.csv")
one = bdb.prepare(trk, plays_bdb, game_id=2022090800, play_id=56)    # flips plays moving left
show_animation(one, times=[-1.0, 0.0, 1.5, 3.0])
# BDB keys are the league's own game and play IDs, so the pbp row (EPA, WP) joins on
# old_game_id == gameId and play_id == playId. nflverse stores old_game_id as text:
# cast it (pbp.old_game_id.astype(int)) before the merge. This play is 2022_01_BUF_LA,
# play 56 in nflverse: the opening snap, Josh Allen's 6-yard completion to Stefon Diggs.
```

Watch for it on Sunday: the Sunday notebook

You have kept a prediction log since How to Watch a Broadcast: before each snap, a run or pass call with a confidence and the clue you used. Scoring it by hand means writing down every result as well. With nflverse you don't have to: the data fills in the results the next morning.

During the season, nflverse refreshes the current season's play-by-play overnight (its README says "nightly"),¹³ so Sunday's games are normally in the file by Monday morning and Monday night's game a day later. Check before you score: print the game IDs you loaded and look for yours. Your weekly routine looks like this (not run here, because the book is built from completed seasons):

```
import nflreadpy as nfl
from nflreadpy.config import update_config

update_config(cache_mode="filesystem", cache_duration=6 * 60 * 60)    # refresh after 6 hours
season = nfl.get_current_season()
week_pbp = nfl.load_pbp([season]).to_pandas()
week_pbp = week_pbp[week_pbp.week == week_pbp.week.max()]    # the latest week in the file
print(sorted(week_pbp.game_id.unique()))    # is your game in yet?
```

Then join your log to the plays and score it. Here is a worked example on Super Bowl LX, with an example log of fifteen Seattle snaps from the first quarter and the start of the second (Seattle's last three snaps of the first quarter are missing, as they would be if you went for a drink), written the way you would write it from the couch: the clock as you read it off the scorebug (often a few seconds off), the down and distance, your call and your confidence. The join can't rely on an exact clock, so it matches each line to the play with the same quarter, down and distance and the **nearest** clock reading, within 15 seconds:

```
log = pd.DataFrame(
    [(1, "15:00", 1, 10, "R", .60, "sit"), (1, "14:31", 1, 10, "P", .55, "align"),
     (1, "13:52", 2, 10, "P", .65, "sit"), (1, "13:27", 1, 10, "R", .55, "sit"),
     (1, "13:00", 1, 10, "P", .60, "align"), (1, "12:53", 2, 10, "R", .55, "align"),
     (1, "12:08", 3, 7, "P", .85, "sit"), (1, "07:13", 1, 10, "R", .60, "sit"),
     (1, "07:08", 2, 10, "P", .65, "align"), (1, "06:24", 3, 6, "P", .85, "sit"),
     (1, "03:31", 1, 10, "R", .65, "align"), (1, "02:53", 2, 8, "R", .55, "align"),
     (1, "02:09", 3, 7, "P", .85, "sit"), (2, "14:12", 1, 10, "R", .55, "sit"),
     (2, "14:06", 2, 10, "P", .60, "sit")],
    columns=["qtr", "clock", "down", "togo", "call", "conf", "clue"])

def secs(mmss):
```

¹³nflfastR README ("updated nightly during the season"): <https://github.com/nflverse/nflfastR>; checked October 2026.

```

m, s = mmss.split(":")
return int(m) * 60 + int(s)

# On a Sunday, swap df25 for week_pbp and put in your game_id and team
game = df25[(df25.game_id == "2025_22_SEA_NE") & (df25.posteam == "SEA")
            & df25.play_type.isin(["pass", "run"]) & df25.down.notna()].copy()
game["t"] = game.time.map(secs)
game["result"] = np.where(game["pass"] == 1, "P", "R") # sacks and scrambles are passes
log["t"] = log.clock.map(secs)
scored = pd.merge_asof(
    log.astype({"qtr": float, "down": float, "togo": float}).sort_values("t"),
    game.rename(columns={"ydstogo": "togo"}).sort_values("t")[["qtr", "down", "togo", "t",
                                                                "result", "epa", "desc"]],
    on="t", by=["qtr", "down", "togo"], direction="nearest", tolerance=15)
scored = scored.sort_values(["qtr", "t"], ascending=[True, False])
scored["hit"] = scored.call == scored.result
# The down-and-distance guesser from 01-06: run on short 1st/2nd/4th downs, pass otherwise
dd_run = ((scored.togo <= 3) & (scored.down != 3)) | ((scored.down == 1) & (scored.togo <= 6))
scored["dd_hit"] = scored.result == np.where(dd_run, "R", "P")
print(f"matched {scored.result.notna().sum()} of {len(scored)} log lines")
print(f"You: {scored.hit.mean():.0%}    always pass: {(scored.result == 'P').mean():.0%}    "
      f"down & distance only: {scored.dd_hit.mean():.0%}")
md_table(scored[["qtr", "clock", "down", "togo", "call", "result", "hit", "clue"]
            .astype({"down": int, "togo": int, "hit": str})])

```

```

matched 15 of 15 log lines
You: 67%    always pass: 60%    down & distance only: 60%

```

qtr	clock	down	togo	call	result	hit	clue
1	15:00	1	10	R	R	True	sit
1	14:31	1	10	P	R	False	align
1	13:52	2	10	P	P	True	sit
1	13:27	1	10	R	P	False	sit
1	13:00	1	10	P	P	True	align
1	12:53	2	10	R	R	True	align
1	12:08	3	7	P	P	True	sit
1	07:13	1	10	R	P	False	sit
1	07:08	2	10	P	P	True	align
1	06:24	3	6	P	P	True	sit
1	03:31	1	10	R	R	True	align
1	02:53	2	8	R	R	True	align
1	02:09	3	7	P	P	True	sit
2	14:12	1	10	R	P	False	sit
2	14:06	2	10	P	R	False	sit

Ten of fifteen, against nine for the down-and-distance guesser and nine for “always pass.” Fifteen snaps prove nothing, of course; Expected Points, EPA, and Success Rate shows how many plays it takes before a rate means anything, and Film Study: All-22, Charting a Game, and Scoring Yourself scores a season’s log properly, with the confidence column. But the habit is the thing: the log you keep on Sunday is a table, and the results arrive as a table on Monday.

💡 Watch for it: the Sunday notebook

- **During the game:** keep the log exactly as in 01-06 (quarter, clock, down, distance, call, confidence, clue). Write the clock as the scorebug shows it just before the snap; being a few seconds off is fine, because the join looks for the nearest play.
- **Monday morning:** load the newest week of play-by-play with a short cache, filter to the game and the offense you watched, and run the join above. Check the match count first: a log line that finds no play usually has a mistyped down or distance.
- **Score it three ways:** your hit rate, “always pass,” and the down-and-distance guesser. Beat both, and you are reading something the scorebug doesn’t show.
- **Read your misses.** Sort the misses by your confidence and read their `desc` text. Confident misses on first-and-10 usually mean you missed a clue in the formation; misses on third down usually mean a draw or a screen you didn’t see coming.
- **Keep the season in one file.** Append each week’s scored log to a CSV. By midseason you can group by `clue` and see which of your reads actually work.

Predict the play

These drills ask you to predict the *data* as well as the football: what the row will say, and what your filters will do with it. The diagrams use the course’s position letters: on offense Q is the quarterback, R the running back, X, Z and H wide receivers, Y and U tight ends; on defense E is an end, T a tackle, S (Sam), M (Mike) and W (Will) the linebackers, C a cornerback, \$ the nickel back, and FS and SS the safeties.

💡 Predict the play 1: write the row

Third-and-2 at the opponent's 40, second quarter, 2025. The offense brings two tight ends and lines up under center; the defense answers with its base 4-3 and one deep safety. The back takes the handoff and gains **3 yards**. Before you look: what will this play's row say for `play_type`, `pass`, `rush`, `down`, `ydstogo`, `yardline_100`, `yards_gained` and `first_down`? What will the next snap's `down`, `ydstogo` and `yardline_100` be? And what will participation's `offense_personnel` string say?

```
off = offense("ace")
dfn = [d.moved(d=5.5, w=-6.5) if d.key == "SS" else d.moved(d=5.0) if d.key in ("LCB", "RCB")
      else d for d in defense("4-3_over", "single_high", off)]
p = Play(off, dfn, los=70, to_go=2, title="3rd & 2 at the opponent's 40")
p.highlight("RB")
p.highlight("Y", "U", color=style.LINE_TO_GAIN)
drop_numbers(p.draw(lateral=(-19, 19), window=(-9, 15)).ax)
plt.show()
```

3rd & 2 at the opponent's 40

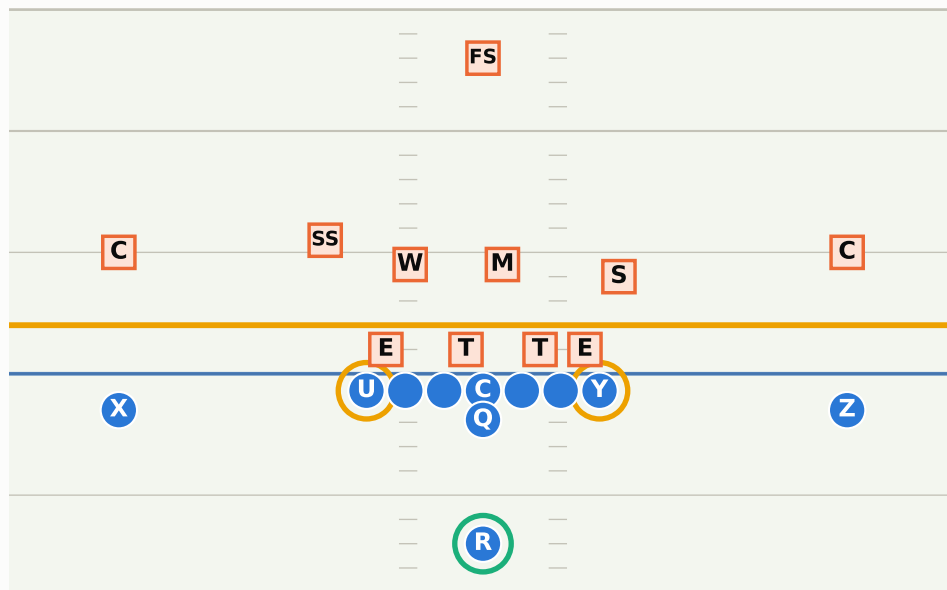


Figure 13: Drill 1: third-and-2 at the opponent's 40. 12 personnel (tight ends Y and U, gold rings) in an under-center 'ace' set against a base 4-3 with one deep safety: the Sam over Y, the strong safety down outside U, the corners pressed up to 5 yards.

The running back (R, aqua ring) will carry for 3 yards. The yellow line is the line to gain, 2 yards ahead.

Answer 1 is at the end of the chapter. Try it first.

💡 Predict the play 2: the scramble

Third-and-8 at your own 35. The offense is in the shotgun with three receivers; the defense plays two-high man, every receiver covered. The quarterback finds nothing, escapes to his right and runs for **9 yards** and a first down. How will nflverse file this play? Will it count as a pass under `play_type == "pass"`? Under `pass == 1`? Which should a pass-rate chart use?

```
off = [q.moved(w=-2.3) if q.key == "RB" else q for q in offense("gun_2x2")]
dfn = [d.moved(d=5.0, w=-6.5) if d.key == "WILL" else d for d in defense("nickel", "two_high", off)]
p = Play(off, dfn, los=45, to_go=8, title="3rd & 8 at your own 35")
p.route("X", [(0, 0), (18, 0)])
p.route("Z", [(0, 0), (18, 0)])
p.route("H", [(0, 0), (10.5, 0), (10.5, 2.5)])
p.route("Y", [(0, 0), (10.5, 0), (10.5, 2.5)])
p.block("RB", pts=[(1.5, 1.5)])
for dfd, rec in [("LCB", "X"), ("RCB", "Z"), ("NB", "Y"), ("WILL", "H"), ("MIKE", "RB")]:
    p.man(dfd, rec)
p.path("QB", [(-2.0, 0.0), (-1.5, 4.5), (14.0, 5.5)])
p.highlight("QB")
p.note("scramble: +9", at=(12.0, 0.5))
drop_numbers(p.draw(lateral=(-19, 19), window=(-9, 15)).ax)
plt.show()
```

3rd & 8 at your own 35

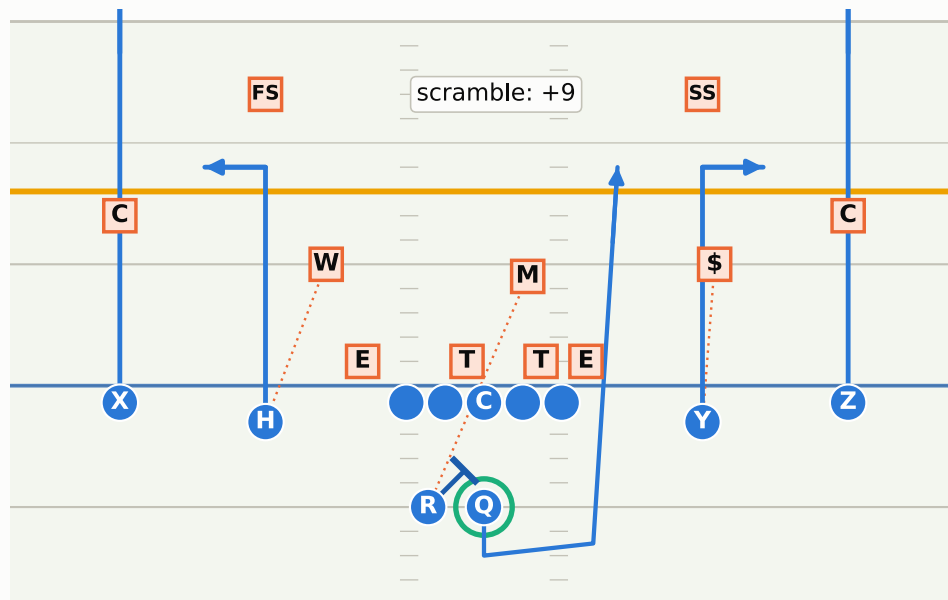


Figure 14: Drill 2: third-and-8 at the offense's own 35. Shotgun 11 personnel against 2-Man: each receiver and the back has a man defender (dotted lines; the Will walked out over the slot H) and two safeties split the deep field. The outside receivers clear out on go routes and the inside receivers run out-breaking routes just past the sticks, all covered, while the back stays in to block. The quarterback (Q, aqua ring) slides right and runs past the yellow line for 9 yards.

Answer 2 is at the end of the chapter. Try it first.

💡 Predict the play 3: called back

First-and-10 at your own 25. The offense runs inside zone to the right and the back breaks it off right tackle for **12 yards**, but there is a flag at the left end of the line: offensive holding on the left tackle, accepted. In nflverse, what is this play's `play_type`, and will your "real plays" filter keep it? What sign will its EPA have? Would the nflfastR guide's filter, `pass == 1 | rush == 1`, keep it? And would your answer change if the hold had been by a receiver 15 yards downfield?

```
off = offense("singleback")
MOVE = {"SDE": (1.0, 5.8), "SAM": (4.2, 7.5), "WDE": (1.3, -4.9), "SS": (8.5, 8.5)}
dfn = [d.moved(*MOVE[d.key]) if d.key in MOVE else d
        for d in defense("4-3_over", "single_high", off)]
p = Play(off, dfn, los=35, to_go=10, title="1st & 10 at your own 25")
p.handoff("RB", t=0.8)
p.run("RB", [(4.5, 2.4), (7.5, 4.0), (11.5, 4.4), (19.0, 5.0)])
for blk, tgt in [("Y", "SDE"), ("RG", "SDT"), ("RT", "MIKE"), ("C", "WDT"), ("LG", "WILL"),
                ("LT", "WDE")]:
    p.block(blk, target=tgt)
p.highlight("RB")
p.highlight("LT", color=style.DEFENSE)
p.note("+12 yards", at=(12.0, 9.5))
p.note("flag: holding (LT)", at=(-4.5, -9.5))
drop_numbers(p.draw(lateral=(-19, 19), window=(-9, 16)).ax)
plt.show()
```

1st & 10 at your own 25

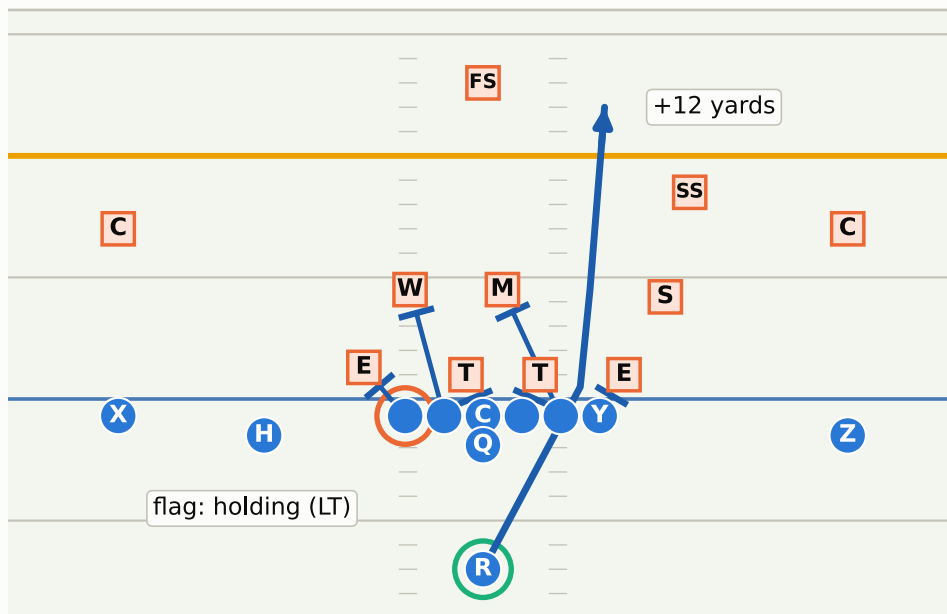


Figure 15: Drill 3: first-and-10 at the offense's own 25. An under-center 11-personnel set against a 4-3 with a wide end outside the tight end. Inside zone to the right: the tight end blocks the end, the right guard the tackle, the right tackle and left guard climb to the Mike and the Will, the center takes the nose, and the left tackle (orange ring) cuts off the backside end, the hardest block on the play to make cleanly, which is why holds happen there. The running back (R, aqua ring) hits the gap outside the right tackle for 12 yards; the flag is on the left tackle, away from the play.

Answer 3 is at the end of the chapter. Try it first.

Takeaways

- **nflverse is free, public, reproducible football data:** play-by-play for every play since 1999, plus schedules, rosters, participation and FTN charting, with nflfastR’s models (EP, EPA, WP, CPOE, xpass) on every row. In Python you load it with **nflreadpy** (Polars; `.to_pandas()` when you want pandas), not the archived `nfl_data_py`.
- **A row is a scorebug plus a result plus model estimates**, and not every row is a play. Situation columns (`down`, `ydstogo`, `yardline_100`) describe the moment before the snap; `yardline_100` is the yards still to go for a touchdown, which is why it falls as an offense advances.
- **Choose rows with `play_type`, classify them with `pass == 1`**, which includes sacks and scrambles, and decide on purpose what to do with no-play penalties, downfield holds, kneel-downs, two-point tries and garbage time. Use win probability 10–90% (outside the last two minutes of each half) by default and for efficiency, 20–80% for comparing play-callers. State your filter every time.
- **Participation and FTN charting add who was on the field and how they lined up**, but participation comes from two suppliers, with a source break between 2022 and 2023: the formation labels, personnel strings and coverage calls all change. Mark the break, translate to what both sides can say (three NGS labels become “under center”), and cross-check against a source that doesn’t break. During the season there is no personnel data at all.
- **Reproduce before you extend.** Rebuilding 02-01’s personnel chart and 01-02’s pass grid to the decimal proves the pipeline; the same dozen lines then answer new questions, such as 27 seasons of pass rate by down.
- **Data can’t tell you where players stood.** A diagram built from a row is part data, part illustration, and the caption must say which; real positions need tracking data.

Connections

Builds on: Downs, Distance, and Field Position (`down`, `distance`, `field position` and the EPA primer, whose pass-rate grid we rebuilt) and Personnel (personnel groupings and the season-by-season chart). The prediction log comes from How to Watch a Broadcast; the formation vocabulary (`shotgun`, `under center`, `pistol`) from Formation Rules and Vocabulary.

Sets up:

- Expected Points, EPA, and Success Rate: builds the `ep` and `epa` columns from scratch with this chapter’s data and filters.
- Win Probability and Decision Analysis: the `wp` column, and the fourth-down decisions behind the chart above.
- Tendencies: PROE, Personnel, Motion, and Coverage: `xpass` and `pass_oe`, neutral situations, and coverage charts that respect the source break.
- Tracking Data and the Big Data Bowl: the real positions that a play-by-play row can’t give you, loaded with `gridiron.bdb`.
- Film Study: All-22, Charting a Game, and Scoring Yourself: the season’s prediction log, scored properly.

If you want to go deeper: the nflreadr data dictionaries for play-by-play, participation and FTN charting (the column definitions, straight from the source); nflfastR’s “A beginner’s guide to nflfastR,” written in R but easy to translate, for the standard filters; Ben Baldwin’s Open Source Football posts for how the models are built; and the nflverse Discord and GitHub issues, where the maintainers answer questions about data quirks.

i Go deeper: a checklist for any new nflverse question

1. **Which table, which seasons?** Check the timeline above before you promise a 2016–2025 chart of something that starts in 2018.
2. **Which rows?** Regular season? `pass == 1` or `play_type`? Penalty snaps? Two-point tries? Garbage time?
3. **Which key?** `pbp game_id + play_id` match participation’s `nflverse_game_id + play_id` and FTN’s `nflverse_game_id + nflverse_play_id`; `gsis_id` for players; `old_game_id` for the Big Data Bowl. Print the match rate after every join.
4. **Does a known total come out right?** A player’s yards, a team’s record, a published league rate.
5. **Does the series cross a source break?** Mark it, translate it, or cross-check it.
6. **Say all of the above in the caption.**

Answers to Predict the play

Answer 1.

The row records the situation **before** the snap and the result in separate columns: `play_type = run`, `pass = 0`, `rush = 1`, `down = 3`, `ydstogo = 2`, `yardline_100 = 40`, `yards_gained = 3`, `first_down = 1`. The next snap is a new series: `down = 1`, `ydstogo = 10`,

`yardline_100 = 37`, because the ball moved 3 yards *closer* to the goal line and `yardline_100` counts down as an offense advances. Say “the next snap”, not “the next row”: a timeout or the end of the quarter can sit in between, so in code you take the next row *that has a down*. The EPA is positive: a third down became a fresh first down.

The personnel string is in FTN’s 2025 format, every player by roster position: “1 C, 2 G, 1 QB, 1 RB, 2 T, 2 TE, 2 WR”. Before 2023 the same play would read “1 RB, 2 TE, 2 WR”. Either way the parser above returns 12 personnel.

One more trap: `yardline_100` falls as the offense gains ground, so “plays inside the 20” is `yardline_100 <= 20`, not `>= 80`, and where a play *ended* is `yardline_100 - yards_gained` only for an ordinary run or completion (a turnover or a penalty breaks it, which is why nflverse also stores `end_yard_line`).

Answer 2.

nflverse files it as `play_type = run`, because the quarterback carried the ball, with `qb_scramble = 1`, `qb_dropback = 1`, `pass = 1` and `rush = 0`. So `play_type == “pass”` counts it as a run, and `pass == 1` counts it as a pass. A chart about **play-calling** should use `pass == 1`: the offense called a pass, and the defense defended one. Over the 2025 regular season that is 1,089 plays (the scramble row of the flags table), and the league’s pass rate on runs and dropbacks was 60.1% by the flag against 56.8% by `play_type`. A chart about **who carried the ball**, such as a quarterback’s rushing yards, is different: there the scramble yards are rushing yards, as the box score says.

The football half: 2-Man is the coverage most exposed to this play. Every underneath defender is running with a receiver, his back to the quarterback, so when the quarterback breaks the pocket nobody is facing him. That is why defenses that play man against a mobile quarterback add a spy or tell the rushers to keep him contained.

Answer 3.

The play never happened, officially. nflverse records it as `play_type = no_play` with `penalty_type = “Offensive Holding”`, `yards_gained = 0` and `rush = 1` (the run is still flagged). The course’s filter, `play_type` in pass or run, **drops** it; `pass == 1 | rush == 1` **keeps** it. Its EPA is clearly **negative**: instead of first-and-10 at the 37, the offense faces first-and-20 at its own 15 (a foul at the line on a run is enforced from the previous spot). In the 2025 regular season there were 176 holds like this one that wiped out runs, and they cost –1.12 expected points each on average.

A hold downfield is a different row. When a receiver, or a lineman who has climbed to the second level, holds beyond the line of scrimmage, the foul is enforced from the spot of the foul, so the run counts up to that spot: the row stays `play_type == “run”` with `penalty = 1`, and your filter **keeps** it. That was 62 of the 238 accepted offensive holds on designed runs in 2025, about a quarter. If you want “runs without penalties”, filtering on `play_type` is not enough; add `penalty == 0`.

Which filter is right depends on the question. “How good is this team’s running game?” arguably should charge the offense for its holds, which is why nflfastR’s guide keeps penalty snaps for EPA. “How often does this team call a run?” is cleaner without them. Expected Points, EPA, and Success Rate measures the difference: leaving penalty snaps out makes passing look a little worse, because defensive pass interference is mostly a deep-ball outcome. Neither choice is wrong. Not knowing which one you made is.